

FSC Safety Manager Installation Guide

**for use with the Honeywell FSC System
Releases 42x, 5xx and 600**

FS20-500

Implementation
FSC Safety Manager

FSC Safety Manager Installation Guide

for use with the Honeywell FSC System
Releases 42x, 5xx and 600

FS20-500
10/01

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About This Publication

This publication includes both general and detailed installation information for the various major components of the FSC Safety Manager (FSC-SM) for use with the Honeywell FSC system Releases 42x, 5xx and 600. Specifically included are:

- general installation instructions for the FSC Safety Manager (FSC-SM) system,
- detailed installation instructions for installing the FSC Safety Manager Module (FSC-SMM), and
- general installation instructions for the FSC-SM system.

Site planning information is contained in the *TotalPlant Solutions (TPS) System Site Planning Manual*, which is in the *System Site Planning* binder. After installation of the FSC-SM, the checkout procedure can be found in the *FSC Safety Manager Service Manual*, which is in this binder.

This publication supports TotalPlant® Solutions (TPS) System network Releases 53x and 62x. TPS is the evolution of TDC 3000^X.

All references in this manual to "FSC Safety Manager" or "FSC Safety Manager Module" pertain only for use with the Honeywell FSC system.

Table of Contents

SECTION 1 – FSC SAFETY MANAGER SYSTEM INSTALLATION	1
1.1 Section Overview	1
1.2 Equipment Warehousing.....	2
1.3 Move Preparation	3
1.4 Unpacking and Placement	13
SECTION 2 – SYSTEM COMPONENT INSTALLATION	21
2.1 Section Overview	21
2.2 Installation Considerations	22
2.3 Cabinet Installation	23
2.4 Cabling Considerations	24
2.5 Cabling Installation	29
2.6 System Power Connections	34
2.7 System Ground Connections	36
2.8 UCN Connections.....	39
SECTION 3 – SYSTEM COMPONENT CONFIGURATION	43
3.1 Section Overview	43
3.2 System Configuration Overview	44
3.3 Module Configuration	45
SECTION 4 – CONFIGURATION.....	47
4.1 Section Overview	47
4.2 Configuration at FSC User Station	48
4.3 Configuration at TPS Universal Station.....	49

Figures

Figure 1-1	Typical Electronics Room Layout.....	3
Figure 1-2	Location and dimensions of the cabinet bolt-down holes	6
Figure 1-3	FSC-SM cabinet with back access (FTA-T).....	7
Figure 1-4	FSC-SM cabinet with front access (FTA-T)	8
Figure 1-5	FSC-SM cabinet with front and back access (FTA-T)	9
Figure 1-6	FSC-SM cabinet with front access (FTA-E)	10
Figure 1-7	FSC-SM cabinet with back access (FTA-E)	11
Figure 1-8	FSC-SM cabinet with front and back access (FTA-E)	12
Figure 1-8	Placement of equipment onto lumber	14
Figure 1-9	Placement using a mobile lifter	16
Figure 1-10	Lifting equipment by mobile lifter of crane	17
Figure 1-11	Lifting equipment by mobile lifter of crane — part 2.....	18
Figure 1-12	Eyebolt crane method	19
Figure 2-1	FSC-SMM front plate	29
Figure 2-2	Redundant FSC-SM UCN cabling	30
Figure 2-3	Typical routing layout of internal wiring between I/O modules and FTA-T via SICs	31
Figure 2-4	Typical routing layout of internal wiring between I/O modules and FTA-E via SICs	32
Figure 2-5	FSC Safety Manager cabinet power connections	34
Figure 2-6	Connection of the UCN	40
Figure 4-1	US System Status display.....	51
Figure 4-2	US Engineering Main Menu display	52
Figure 4-3	US NIM Build Type Select Menu.....	53
Figure 4-4	US UCN Node Configuration display	54
Figure 4-5	US Node-Specific Configuration display	55
Figure 4-6	US Process Point Building display	56

Tables

Table 1-1	Move preparation checklist	4
Table 1-2	Move preparation checklist – part 2	5
Table 1-3	Important unpacking considerations	13
Table 1-4	Fork lift considerations	14
Table 1-5	Mobile lifter considerations	15
Table 1-6	Roller method considerations	18
Table 2-1	FSC system installation checklist.....	22
Table 2-2	Raceway shielding	24
Table 2-3	IEEE 518 wiring classification definitions	25
Table 2-4	Tray spacing.....	26
Table 2-5	Tray/conduit spacing.....	26
Table 2-6	Conduit spacing	27
Table 2-7	Cabling guidelines	28
Table 2-8	System grounding considerations	36
Table 4-1	Start-up and reconfiguration procedures	49
Table 4-2	UCN configuration.....	50
Table 4-3	UCN node configuration.....	53
Table 4-4	FSC-SMM data point configuration procedure.....	57

Acronyms

AC		Alternating Current
CP		Central Part
CPU		Central Processor Unit
DBM		Diagnostic and Battery Module
DC		Direct Current
DCS		Distributed Control System
EMI		Electromagnetic Interference
EPROM		Erasable Programmable Read-Only Memory
ESD		Electrostatic Discharge
FLD		Functional Logic Diagram
FSC		Fail Safe Control
FSC-DS		FSC Development System
FSC-SM		FSC Safety Manager
FSC-SMM		FSC Safety Manager Module
FTA		Field Termination Assembly
HM		History Module
I/O		Input/Output
IEEE		Institute of Electrical and Electronic Engineers
LCN		Local Control Network
LED		Light Emitting Diode
NCF		Network Configuration File
NIM		Network Interface Module
PLC		Programmable Logic Controller
SIC		System Interconnection Cable
SM		Safety Manager
SMM		Safety Manager Module
TAC		Technical Assistance Center
TPS		TotalPlant Solution
TTL		Transistor - Transistor Logic
UCN		Universal Control Network
US		Universal Station
VBD		Vertical Bus Driver
WD		Watchdog

References

For FSC-SM documentation:

Publication Title	Publication Number	Binder Title	Binder Number
<i>FSC Safety Manager Installation Guide</i>	FS20-500	Implementation FSC Safety Manager	TPS 3076
<i>FSC Safety Manager Implementation Guidelines</i>	FS11-500	Implementation FSC Safety Manager	TPS 3076
<i>FSC Safety Manager Control Functions</i>	FS09-500	Implementation FSC Safety Manager	TPS 3076
<i>FSC Safety Manager Parameter Reference Dictionary</i>	FS09-550	Implementation FSC Safety Manager	TPS 3076
<i>FSC Safety Manager Configuration Forms</i>	FS88-500	Implementation FSC Safety Manager	TPS 3076
<i>FSC Safety Manager Service Manual</i>	FS13-500	Implementation FSC Safety Manager	TPS 3076

For FSC documentation:

Publication Title	Publication Number
<i>FSC Safety Manual</i>	PM.MAN.8047
<i>FSC Hardware Manual</i>	PM.MAN.8048
<i>FSC Software Manual</i>	PM.MAN.8025

Section 1 – FSC Safety Manager System Installation

1.1 Section Overview

About this section

This section provides general installation information for the FSC Safety Manager (FSC-SM) system. Topics included in this section are:

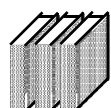
Subsection	Topic	See Page
1.1	Section Overview	1
1.2	Equipment Warehousing	2
1.3	Move Preparation.....	3
1.4	Unpacking and Placement.....	13

Tools and test equipment

The following equipment is necessary for installation of the FSC Safety Manager:

- Mobile lifter/lift truck — refer to Subsection 1.4 (*Unpacking and Placement*),
- Carrier band MAP tester — refer to Subsection 2.2 (*Installation Considerations*),
- Hand tools and multimeter, as in a typical electrician's tool kit,
- UCN Connector Torque Wrench Kit (51109612-100),
- Set of SAE (inch) "crow's foot" wrenches for the above torque wrench,
- ESD kit (Honeywell part number 30185-H).

ATTENTION Installation of the FSC Safety Manager may **only** be carried out by qualified and authorized personnel. Failure to comply with the regulations and recommendations contained in this document may cause severe damage to the equipment or serious injury to people.



REFERENCE — For detailed information regarding

- site planning and installation, refer to SM02-550 *TPS System Site Planning Manual* in Binder TPS 3020-1 *System Site Planning-I*.
- post-installation checkout procedures, refer to the *FSC Safety Manager Service Manual* in this binder.

1.2 Equipment Warehousing

Factory packing

For shipment, the FSC Safety Manager is packed with desiccant packages, wrapped in a moisture barrier and optionally enclosed for shipment as required by the user.

Transit/warehousing specifications

The FSC Safety Manager can be shipped by a variety of means, but any trucking should be done by enclosed air-ride vans. The equipment is usually trucked on the final leg of the journey to the user warehouse or site. There it is stored off the ground on a solid piece or combination of lumber pieces (minimum height 10 cm / 4 in) and indoors until needed for installation. During the whole process, the environment must be monitored and corrected if the following equipment transit/storage ratings are exceeded:

- Temperature range: -20°C to $+70^{\circ}\text{C}$ (-4°F to $+158^{\circ}\text{F}$)
- Relative humidity: 5% to 95%, non-condensing
- Vibration: 1 G, frequency 10-150 Hz (IEC 68-2-6)
- Shock: 15 G for 11 ms (IEC 68-2-27)

ATTENTION In order to minimize exposure to humidity, keep the factory wrapping intact during storage and transit. If it is necessary to unseal the equipment for customs or receiving, add more desiccant and reseal the package.

1.3 Move Preparation

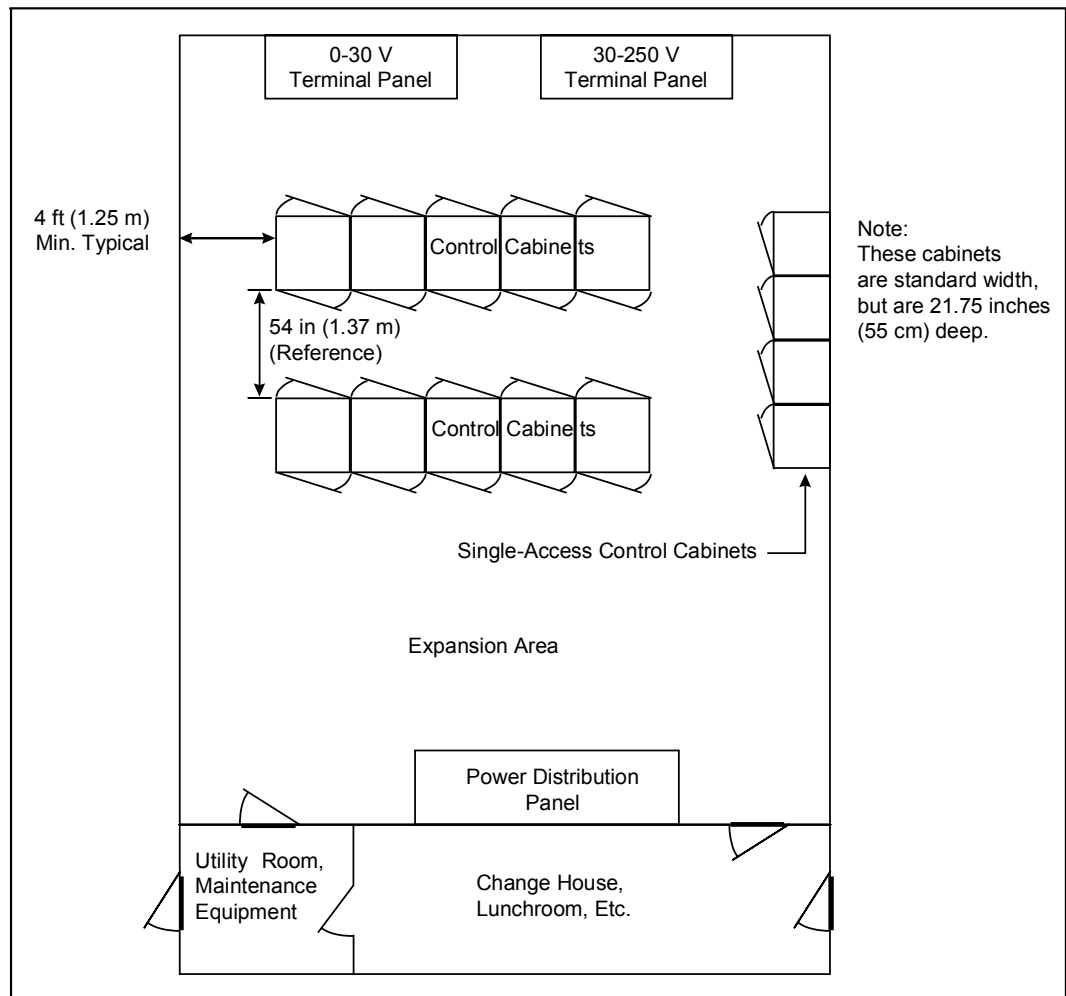
Background

Several basic points need to be kept in mind when planning the movement and unpacking of FSC Safety Manager cabinets. They include the following:

- A single FSC Safety Manager cabinet is nominally 80 x 80 x 200 cm (31½ x 32½ x 78¾ inches),
- Lifting eye bolts, which are removable, add about 5 cm (2 inches) to the height, and
- Weight varies with options and can go as high as 385 kg (850 lb) per cabinet.

As illustrated in Figure 1-1, a typical factory-assembled complex uses up to four such cabinets bolted together.

Figure 1-1 – Typical Electronics Room Layout



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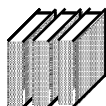
1.3 Move Preparation, Continued

Checklist

Table 1-1 provides a checklist to be used when planning the movement of FSC Safety Manager cabinets.

Table 1-1 – Move preparation checklist

Check	Action	Notes
1	Travel Route	Chart the equipment's route of travel regarding: ☐ loading doors — height and location, ☐ elevators — size, capacity, and availability, ☐ aisles — number, size, and layout, ☐ doorways — number and size, ☐ hallways — size, restrictions, and bends, ☐ stairs and ramps — may require some type of lifting, and ☐ rigging of equipment — if needed.
2	Space	Is there enough space to hold the equipment?
3	Floor Loading	Will the floor in all areas be strong enough?
4	Power Distribution	Check for proper transformer sizes, distribution panels and sufficient outlets for support, test and maintenance equipment.
5	Grounding	Be sure that proper grounds are available.
6	Conduits and Cable Trays	Make sure that conduits and cable trays are large enough and the proper type to accommodate all of the required cables.
7	Environment	Are lighting and other environmental requirements adequate?
8	Security	Provide security clearances for the installation crew if it is not composed of regular employees.
9	Responsibilities	Are all responsibilities clearly defined and understood?
10	Communications	At minimum, three conveniently located telephones are needed for: ☐ Honeywell Technical Assistance Center (TAC) data terminal, ☐ Honeywell TAC voice, and ☐ Operator Emergency.



For additional information regarding actions described in the Move Preparation Checklist, refer to SM02-550 — *TPS System Site Planning Manual* in the Binder TPS 3020-1 *System Site Planning-1*.

1.3 Move Preparation, Continued

Preparation

Table 1-2 provides a checklist to be used when preparing the electronics room for the FSC Safety Manager cabinets.

Table 1-2 – Move preparation checklist – part 2

Check	Action	Notes
1	Static Electricity	Reduce the ESD-borne failures by: □ stripping the floors of any wax to prevent static charge buildup, □ using anti-static spray on floors, furniture and fabric in the work area, and □ placing a sign on or near the equipment that states circuit boards are not to be handled without wearing a ground strap.
2	Tack Rugs	Placed at all entrances, tack rugs catch dust, grit and abrasives from other areas.
3	Fire Prevention	Consult with the local fire prevention authority to select fire extinguishers suitable for electrical fires.
4	Room Layout	Prepare an electronics room layout diagram for cabinet placement. • Refer to Figure 1-1.
5	Cabinet Bolt-Down	If this FSC Safety Manager cabinet is to be bolted down, pre-drill holes in the floor. ATTENTION Refer to Figure 1-2 for placement and dimensions of the bolt-down holes.

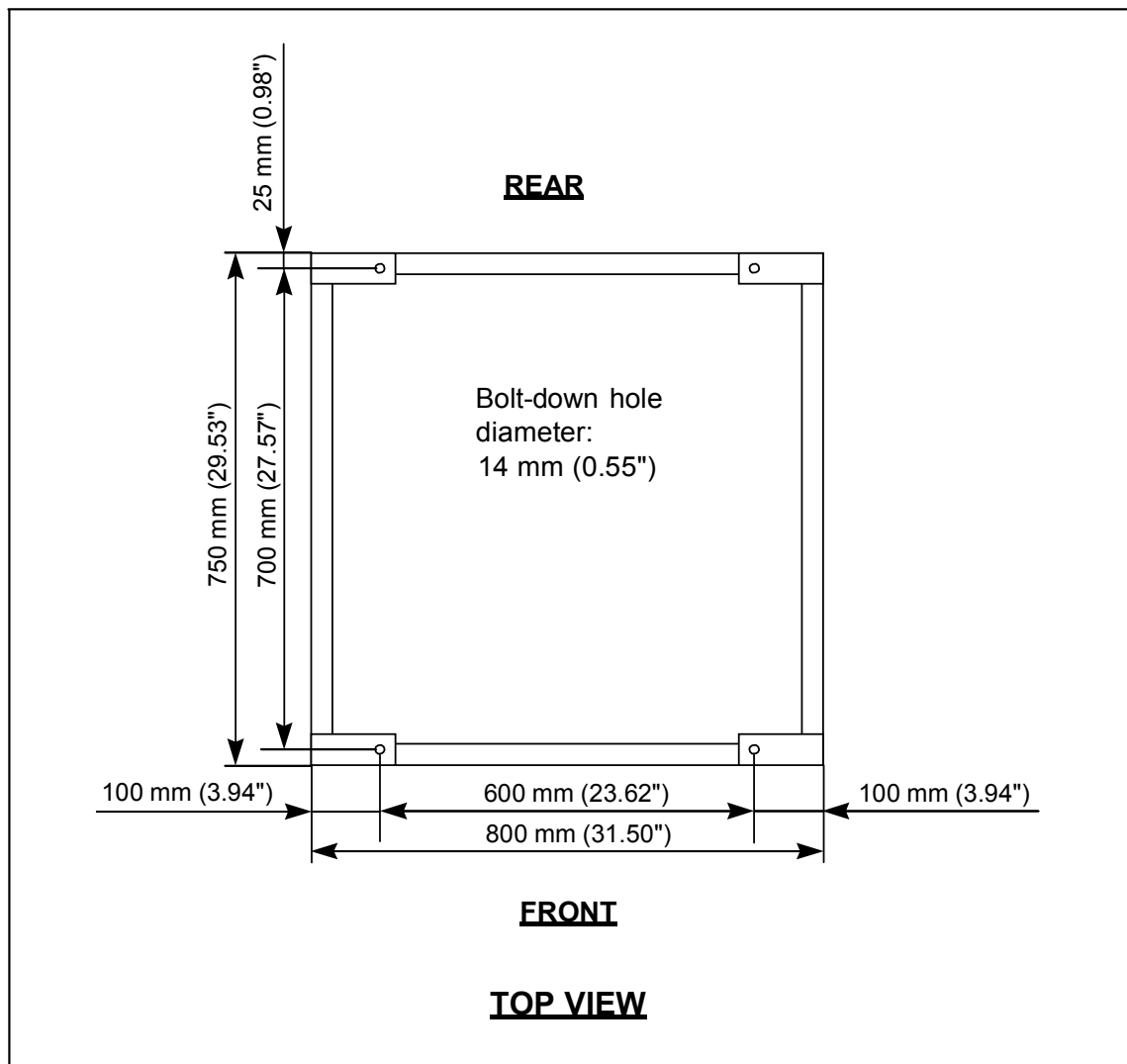
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1.3 Move Preparation, Continued

Bolt-down holes

Figure 1-2 provides the exact locations and dimensions of the cabinet bolt-down holes.

Figure 1-2 – Location and dimensions of the cabinet bolt-down holes



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1.3 Move Preparation, Continued

Cabinet entry

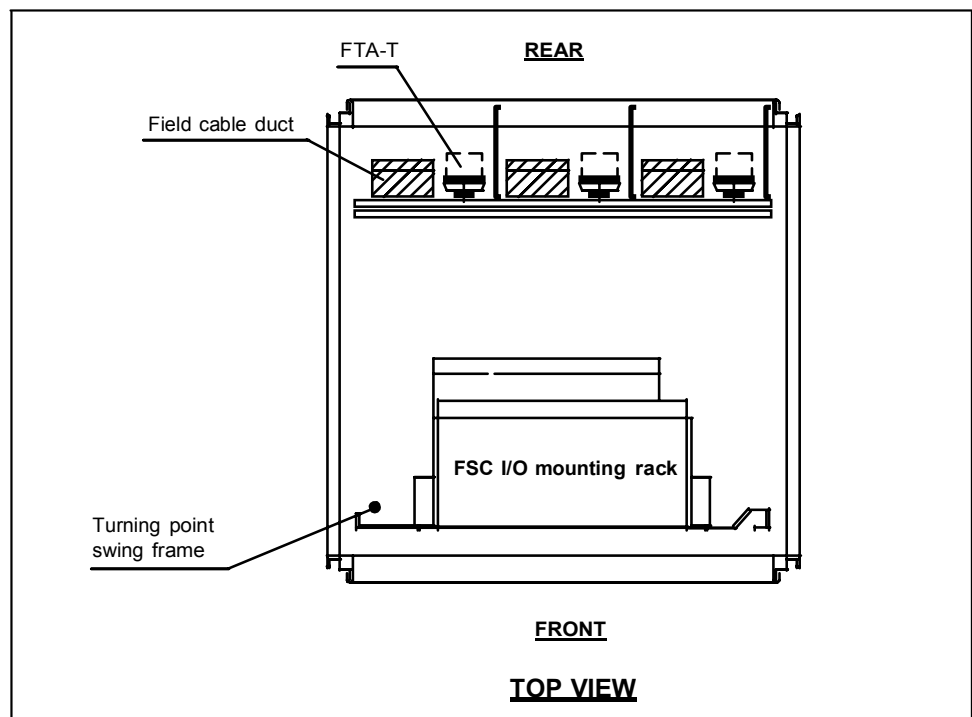
The process signal cables can enter the FSC Safety Manager cabinet through the top or bottom of the cabinet. The FSC-SM cabinet is prepared for field termination assembly to interface boards (FTA-E) or terminals (FTA-T). These interface boards or terminals may be located in the FSC-SM cabinet or in cross-wiring cabinets next to the FSC-SM cabinet. Figures 1-3 through 1-8 illustrate the placement and dimensions for process wiring for FTAs located in the FSC-SM cabinet.

The FSC-SM cabinet is also prepared for top or bottom entry as per user specification. For bottom entry, sliding floor plates are provided which can be adjusted to provide entry slots. For top entry, one sliding plate is provided which can be adjusted to provide entry slots. For top or bottom entry, the cables should be clamped firmly to the cabinet. The clamping devices should be able to withstand a 45 kg (100 lb) pull. This prevents damage inside the cabinet when pulling on cables outside the cabinet.

The FSC-SM cabinet includes a cable clamp rail at the bottom or at the top.

Figure 1-3 illustrates an FSC-SM cabinet bottom configured for back access of the I/O field cables with FTA-T.

Figure 1-3 – FSC-SM cabinet with back access (FTA-T)



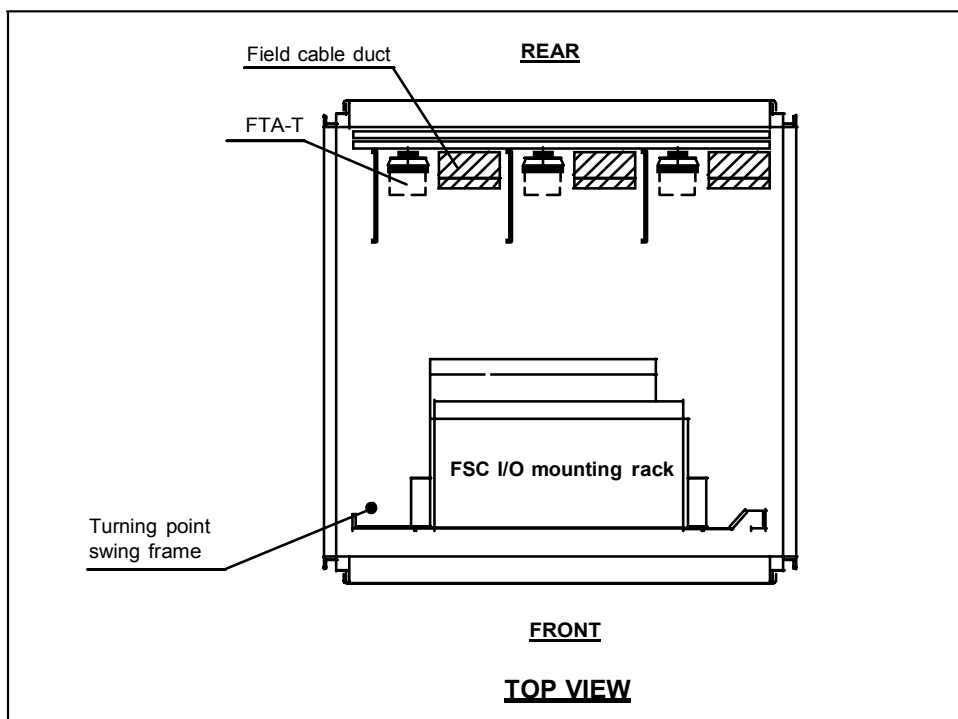
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1.3 Move Preparation, Continued

Cabinet entry,
continued

Figure 1-4 illustrates an FSC Safety Manager cabinet bottom configured for front access of the I/O field cables with FTA-T.

Figure 1-4 – FSC-SM cabinet with front access (FTA-T)



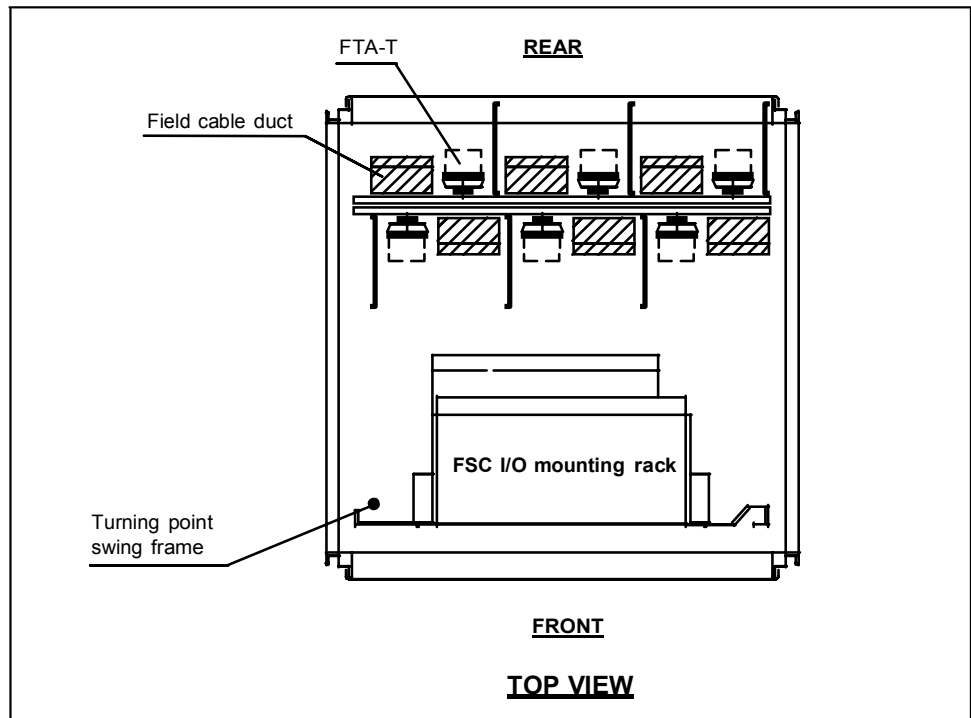
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1.3 Move Preparation, Continued

Cabinet entry, continued

Figure 1-5 illustrates an FSC Safety Manager cabinet bottom configured for front and back access of the I/O field cables with FTA-T.

Figure 1-5 – FSC-SM cabinet with front and back access (FTA-T)



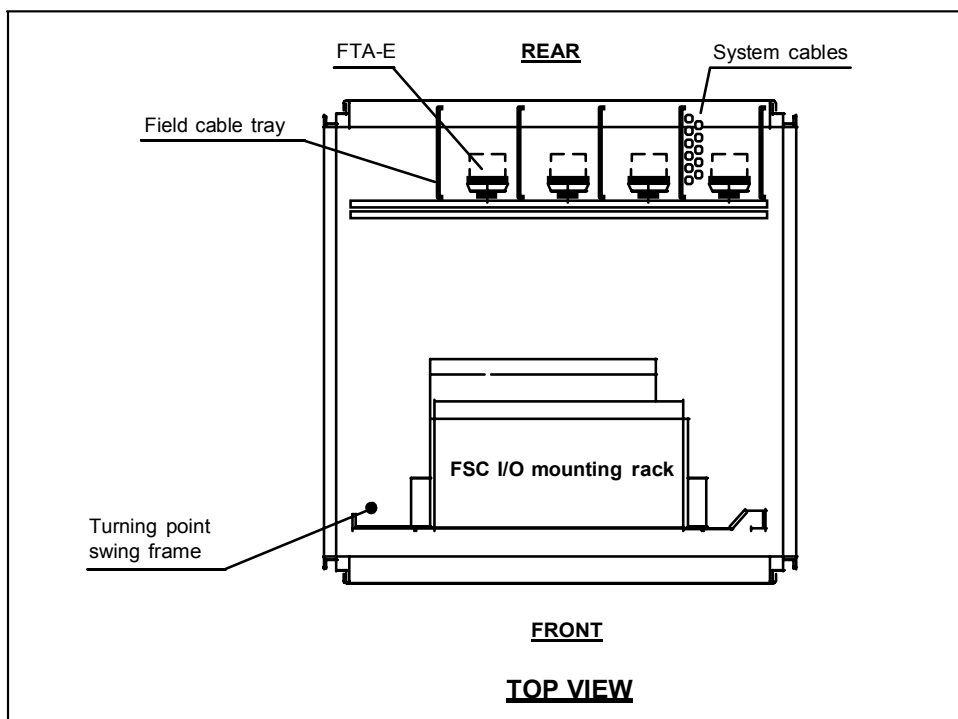
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1.3 Move Preparation, Continued

Cabinet entry, continued

Figure 1-6 illustrates an FSC Safety Manager cabinet bottom configured for front access of the system cables with FTA-E.

Figure 1-6 – FSC-SM cabinet with front access (FTA-E)



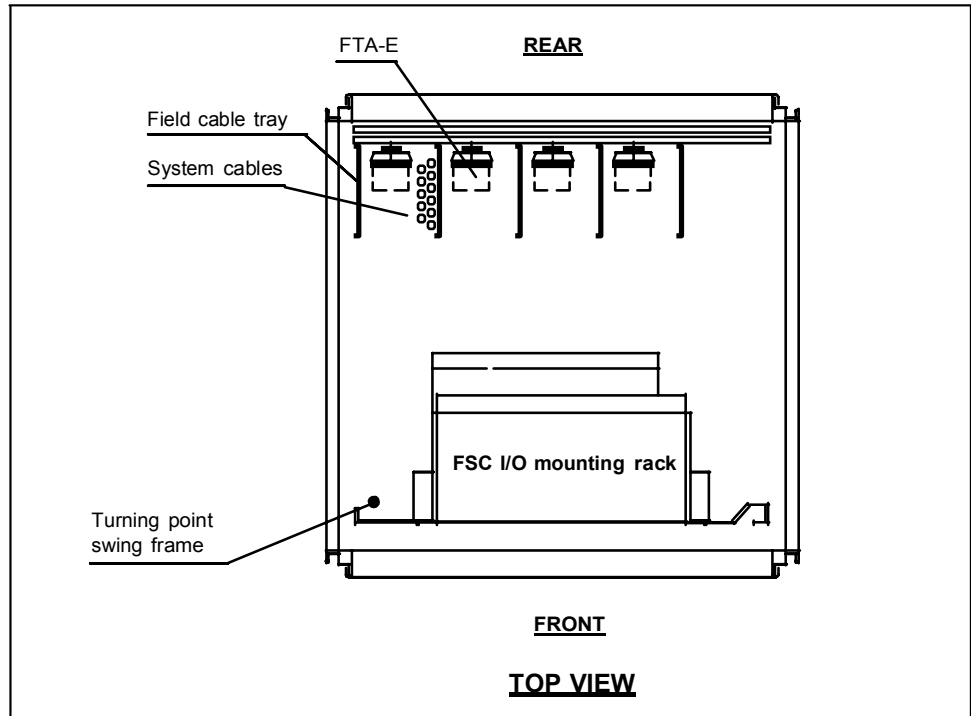
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1.3 Move Preparation, Continued

Cabinet entry, continued

Figure 1-7 illustrates an FSC Safety Manager cabinet bottom configured for back access of the system cables with FTA-E.

Figure 1-7 – FSC-SM cabinet with back access (FTA-E)



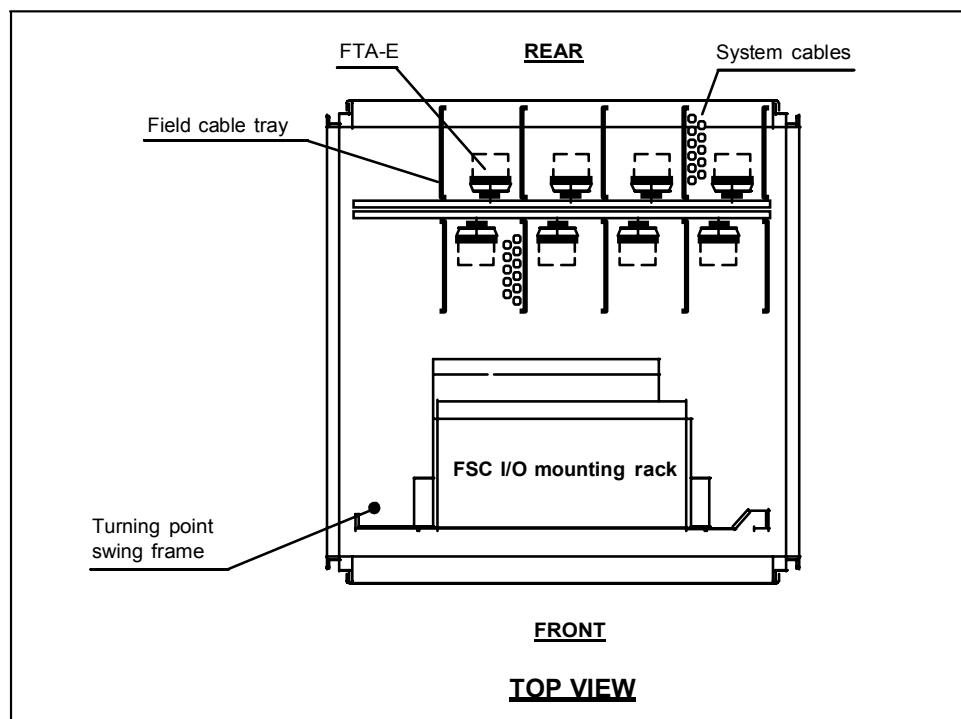
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1.3 Move Preparation, Continued

Cabinet entry, continued

Figure 1-8 illustrates an FSC Safety Manager cabinet bottom configured for front and back access of the system cables with FTA-E.

Figure 1-8 – FSC-SM cabinet with front and back access (FTA-E)



1.4 Unpacking and Placement

Unpacking

Unpack the equipment, check the shipment against the invoice, and immediately notify your Honeywell representative if any items are missing. Table 1-3 lists important considerations when unpacking the equipment.

Table 1-3 – Important unpacking considerations

IF ...	THEN ...
the FSC Safety Manager system is comprised of individual modules	the component parts are packaged in labeled boxes.
the package has visual damage	the equipment inside may also be damaged.
the package is removed and equipment has visual damage	inform the insurance company and your Honeywell representative. In any case, do not operate the equipment until confirmation from Honeywell Safety Management Systems has been obtained.

Moving

The FSC Safety Manager equipment can be moved in any one of four methods:

- fork lift,
- mobile lifter,
- roller, or
- crane.

Special consideration for each of these methods is provided in the following information.

CAUTION

When moving the FSC Safety Manager equipment:

- use lifters with a capacity of 1100 kg (2500 lb) or greater, and
- do not tilt the cabinet more than 45° front or back to avoid damaging the doors (they are not recessed).

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1.4 Unpacking and Placement, Continued

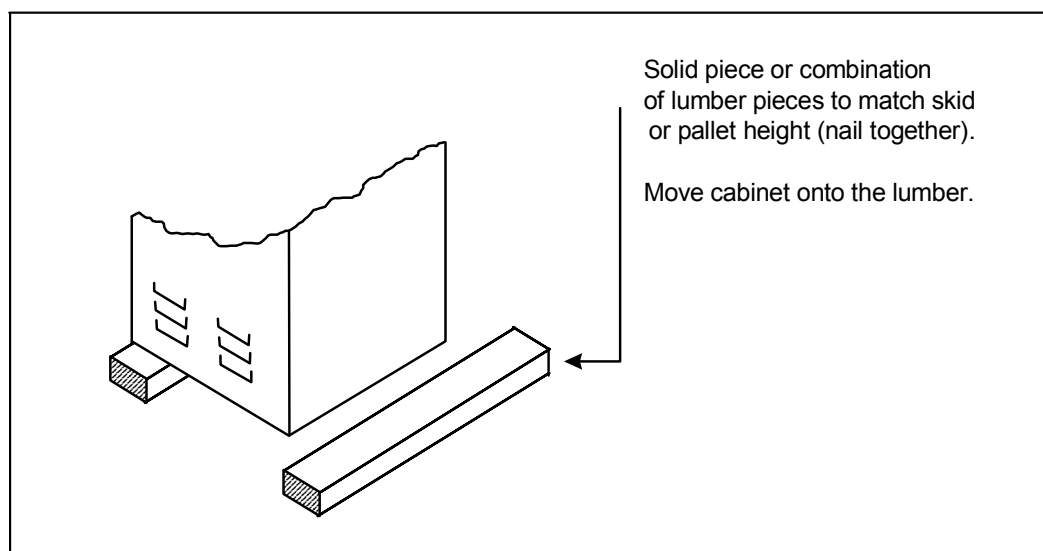
Fork lift

The FSC-SM cabinet is moved by fork lift to the final location. Table 1-4 lists special considerations regarding the use of a fork lift to move the equipment.

Table 1-4 – Fork lift considerations

Number	Consideration
1	Move the equipment to the electronics room by fork lift; remove the fork lift by placing the equipment onto the lumber pieces by 10 cm (4 inches) as illustrated in Figure 1-9.
2	Tip the cabinet off the lumber and onto the floor; remove the lumber.
3	Manually move the cabinet to the exact final position.

Figure 1-9 – Placement of equipment onto lumber



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1.4 Unpacking and Placement, Continued

Mobile lifter

As illustrated in Figure 1-7, the FSC Safety Manager is placed on a transport skid made from angle iron. It is then moved with a mobile lifter. Table 1-5 lists special considerations regarding the use of a mobile lifter to move the equipment.

Table 1-5 – Mobile lifter considerations

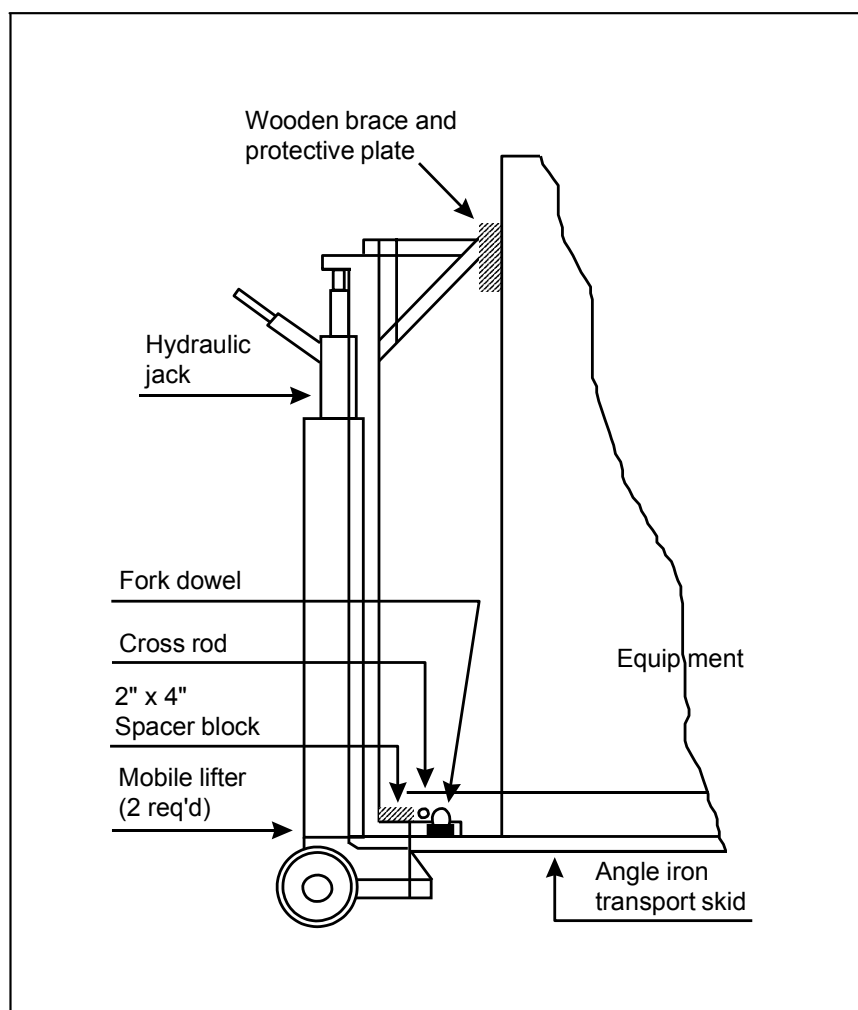
Number	Consideration
1	Tilt the equipment slightly and slide one of the angle irons used to make up a transport skid underneath the cabinet as in Figure 1-10.
2	Repeat the above step for the other angle.
3	Insert a strip of heavy cardboard between each angle iron and the equipment finish.
4	Tighten the angle irons against the cabinet by tapping slightly.
5	Insert and tighten two 12 mm (0.5 in) minimum, threaded cross rods as in Figure 1-10 and Figure 1-11.
6	Spread the lifter forks to 80 cm (31.5 in) centers and put them under each cross rod. Place the fork's dowels between the cross rods and the cabinet, just touching the cross rods.
7	Place wooden spacer blocks along the forks, long enough to press the cross rods against the fork dowels.
8	Position a protective plate using braces as in Figure 1-10. Be sure that the plate does not press against handles or other cabinet hardware.
9	Raise both lifter forks evenly until the equipment clears the shipping skid; pull it out of the way.
10	Wheel the equipment to the final position in the electronics room; lower it to the floor.
11	Pull out the mobile lifters. Dismantle and remove the angle iron transport skid.
12	Manually place the cabinet in the exact final position.

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1.4 Unpacking and Placement, Continued

Mobile lifters, continued

Figure 1-10 – Placement using a mobile lifter

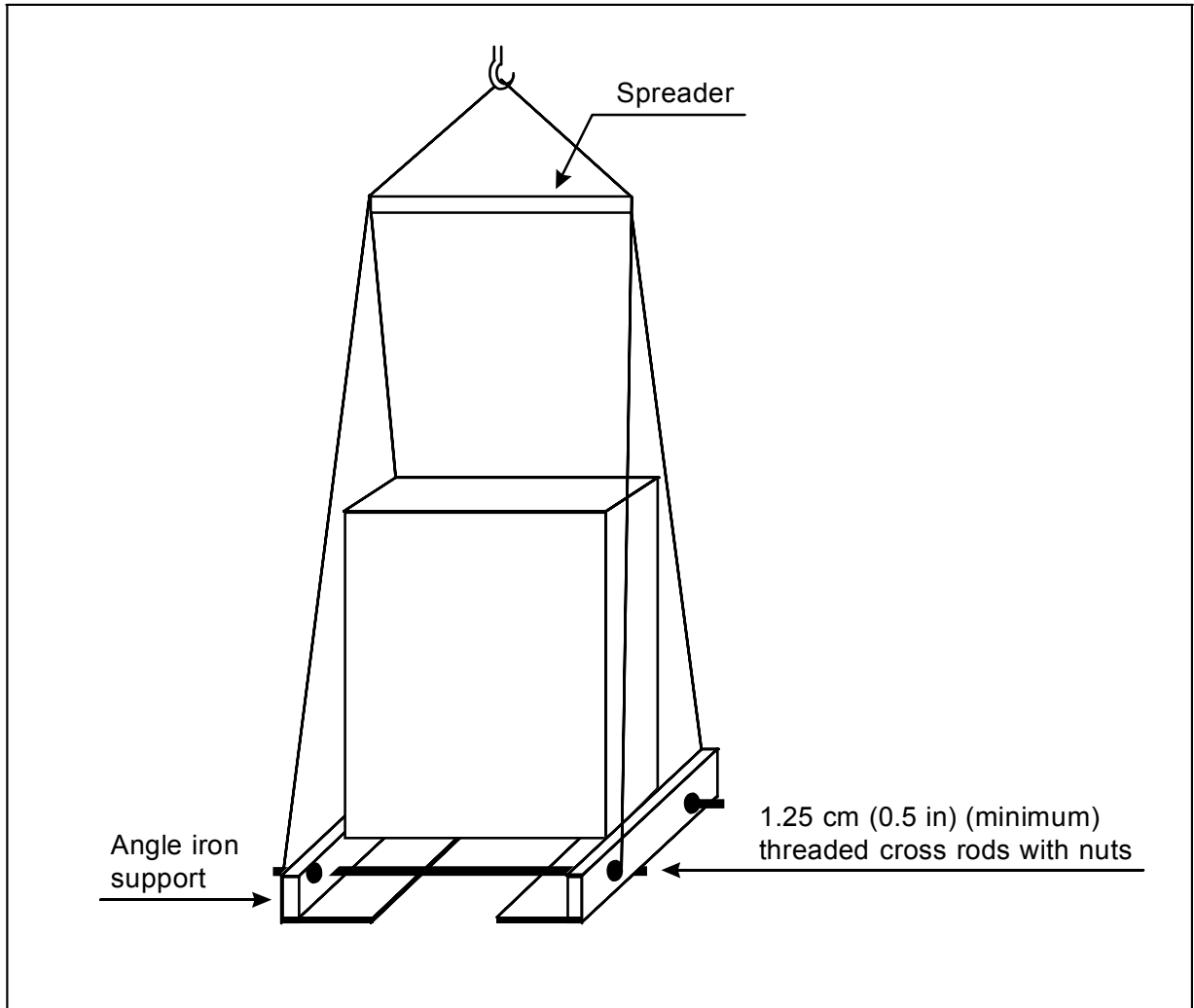


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1.4 Unpacking and Placement, Continued

Mobile lifters, continued

Figure 1-11 – Lifting equipment by mobile lifter of crane



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1.4 Unpacking and Placement, Continued

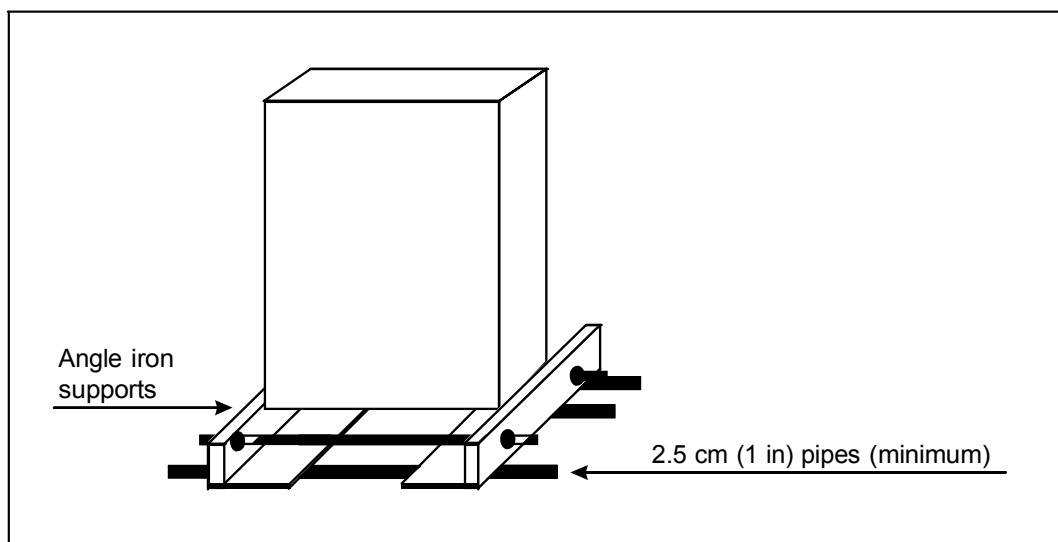
Roller method

As illustrated in Figure 1-12, using the roller method the FSC Safety Manager is moved to its final destination by placing pipe rollers under a transport skid made of angle iron. Table 1-6 lists special considerations regarding the use of the roller method to move the equipment.

Table 1-6 – Roller method considerations

Number	Consideration
1	Use one of the previous methods to move the cabinet to a location where rollers are to be used.
2	Install the angle iron transport skid as in the mobile lifter method.
3	Place 12 mm (1/2 inch) pipes under the transport skid as in Figure 1-12. ATTENTION The load must be on the angle irons, not the cabinet.
4	Roll the cabinet to its final location. ATTENTION At least three pipes must be under the transport skid at all times.
5	When the cabinet is near the final location, tilt it and pull out the pipes, the middle one last.
6	Dismantle the transport skid and position the cabinet manually to the exact final location.

Figure 1-12 – Lifting equipment by mobile lifter of crane — part 2



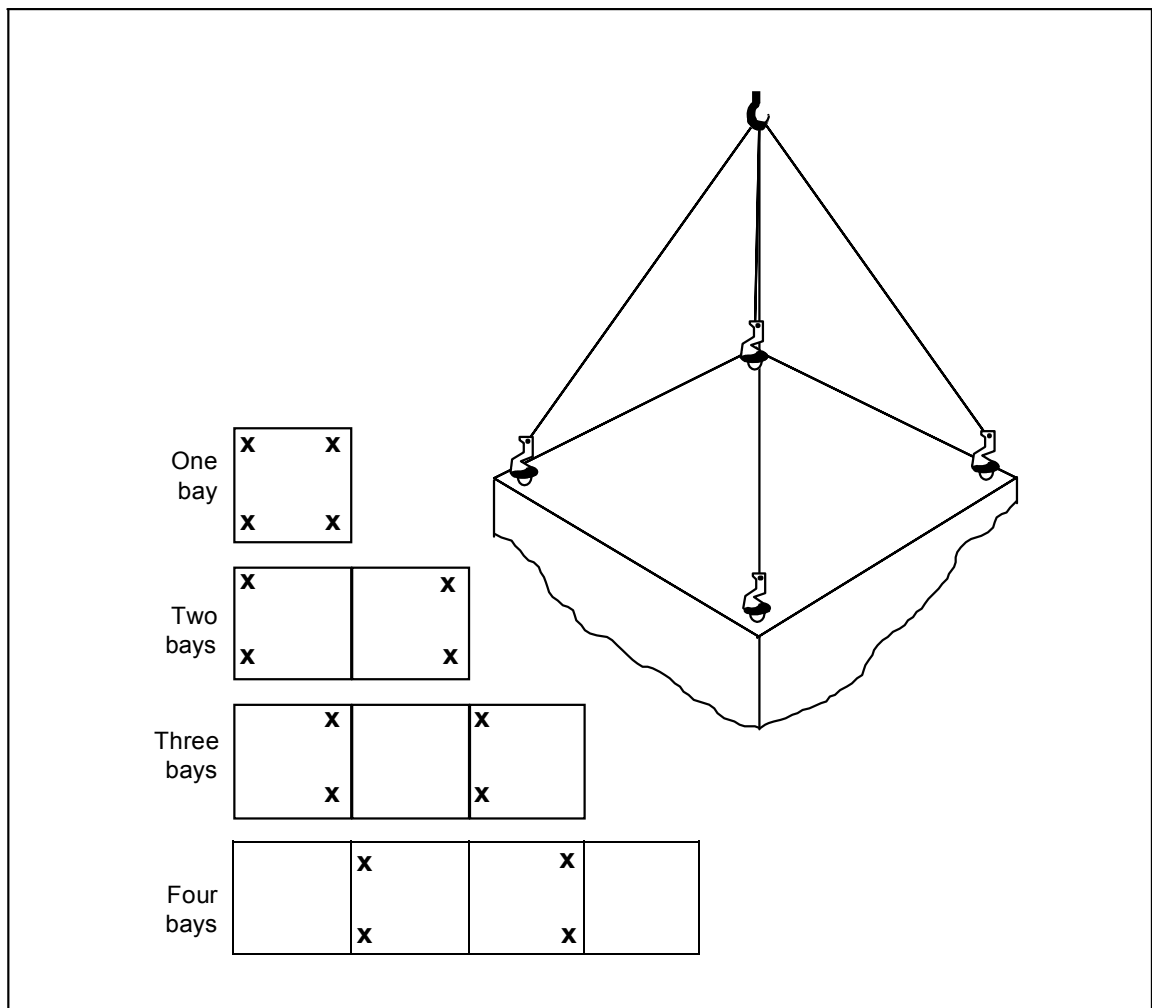
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1.4 Unpacking and Placement, Continued

Crane method

In this method the cabinet is lifted into place with a crane. Use either an angle iron transport skid (see Figure 1-12) or the lifting eyebolts installed on each cabinet (see Figure 1-13).

Figure 1-13 – Eyebolt crane method



Leveling and bolt-down

FSC Safety Manager cabinets do not include levelers; it may be necessary to hand-fit shims underneath the cabinet in order to make the cabinet sit solidly.

The FSC-SM cabinet has four holes in the bottom for bolting it to a floor. A partially filled cabinet should be bolted down. Cabinets that are bolted together are not easily moved and do not ordinarily require bolting down. Refer to Figure 1-2 for bolt-down dimensions.

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Section 2 – System Component Installation

2.1 Section Overview

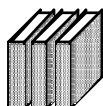
About this section

This section provides general installation information for various components of the FSC Safety Manager (FSC-SM) system. Topics included in this section are:

Subsection	Topic	See Page
2.1	Section Overview	21
2.2	Installation Considerations	22
2.3	Cabinet Installation	23
2.4	Cabling Considerations	24
2.5	Cabling Installation	29
2.6	System Power Connections	34
2.7	System Ground Connections	36
2.8	UCN Connections	39

ATTENTION

This section describes in detail only those installation procedures and considerations specific to the FSC Safety Manager Module (FSC-SMM) and the TotalPlant Solutions (TPS) system. Installation procedures and considerations specific to the FSC components can be found in the FSC documentation.



REFERENCE — For detailed information regarding

- site planning and installation, refer to SM02-550 — *TPS System Site Planning Manual* in Binder TPS 3020-1 *System Site Planning-1*.
- post-installation checkout procedures, refer to FS13-500 — *FSC Safety Manager Service Manual* in this binder.

2.2 Installation Considerations

Introduction

The FSC Safety Manager system may encompass a single cabinet or a multi-cabinet complex, or may be delivered without a cabinet.

If your FSC-SM system includes a cabinet or cabinets, then power, ground and cabling are brought to the system and connected. The system is configured and installation is complete.

Should your FSC-SM system not include a cabinet or cabinets, the individual chassis must be mounted to a structure and power and ground supplied by a choice of methods.

Installation checklist

When installing an FSC-SM system, consider the installation checklist as shown in Table 2-1.

Table 2-1 – FSC system installation checklist

Step	Action	Done (☐)
1	Carefully review all documentation available for the equipment, paying close attention to cautionary notes.	
2	Install the FSC-SM equipment in accordance with the requirements as laid down in the Integration Guidelines for FSC System cabinets.	
3	Use technically qualified personnel to service the FSC-SM components.	

2.3 Cabinet Installation

Introduction

After placement, the cabinet or cabinet complex requires the attachment of power and ground, and cabling installation.

ATTENTION Refer to Section 1 — *FSC Safety Manager System Installation* for information regarding the warehousing, movement, unpacking and placement of the FSC Safety Manager cabinets.

2.4 Cabling Considerations

Introduction

When installing an FSC Safety Manager System, it is important to consider:

- conduit and tray shielding and spacing, and
- cable and wire classifications.

Shielding

Conduit or cable trays should be used to route the network trunk line within the plant for short distances or for several miles. The same conduit should not carry both data and power wiring because of problems created by electromagnetic interference (EMI). The choice of materials and configuration for the cable conduit can determine the degree of protection from this interference.

Table 2-2, showing raceway shielding, is taken from IEEE Standard 518 for the installation of electrical equipment. It compares different types of conduits and raceways for magnetic field attenuation at 60 Hz and for electrostatic attenuation at 100 kHz.

Table 2-2 – Raceway shielding

Raceway Type	Thickness mm (in.)	60 Hz Magnetic Field Attenuation		100 kHz Magnetic Field Attenuation	
		Ratio	dB	Ratio	dB
Free air		1:1	0	1:1	0
2 inch (5 cm) aluminum conduit	3.91 (0.154)	1.5:1	3.3	2.150:1	66.5
#16 gauge aluminum tray	1.52 (0.060)	1.6:1	4.1	15.500:1	83.9
#16 gauge aluminum tray	1.52 (0.060)	3:1	9.4	20.000:1	86.0
#16 gauge galvan. iron tray	1.52 (0.060)	3.2:1	10.0	22.000:1	86.8
2 inch (5 cm) IPS copper pipe	3.96 (0.156)	3.3:1	10.2	10.750:1	80.6
#16 gauge aluminum tray	1.52 (0.060)	4.2:1	11.5	29.000:1	89.3
#14 gauge aluminum tray	1.90 (0.075)	6:1	15.5	23.750:1	87.5
2 inch (5 cm) metallic tubing	1.65 (0.065)	6.7:1	16.5	3.350:1	70.5
2 inch (5 cm) rigid galvanized conduit	3.91 (0.154)	40:1	32.0	8.850:1	78.9

Continued on next page

2.4 Cabling Considerations, Continued

Spacing

It is recommended that noise creating sources (those with time varying voltage or time varying current) be separated from data signal cable. Cables with similar levels of noise susceptibility should be grouped together, and those with similar levels of noise generation should be separately grouped in trays and conduits.

IEEE 518 defines four classes of wiring that differ in signal level and noise susceptibility as provided in Table 2-3.

Table 2-3 – IEEE 518 wiring classification definitions

Class	Definition
Data 1	High noise susceptibility. Analog signals of less than 50 V and digital signals of less than 15 V.
Data 2	Medium susceptibility. Analog signals greater than 50 V, regulating signals of 50 V with currents less than 20 A; AC feeders less than 20 A.
Data 3	Low susceptibility. Switching signals greater than 50 V, analog signals of 50 V with currents less than 20 A; AC feeders less than 20 A.
Data 4	Power AC and DC buses of 0-1000 V with currents of 20-800 A.

ATTENTION Data 3 and Data 4 classes have one subclass each for applications that require special cable and spacing. Table 2-4, Table 2-5 and Table 2-6 in this section list the subclass for Data 3—listed as Data 3A; the subclass for Data 4 is listed as Data 4A. Examples of Data 3A and Data 4A subclass cables include:

- signals from communicating fields and line resistors,
- signals from shunts to regulators, and
- power greater than 1000 volts or greater than 800 amperes, or both.

Table 2-4, Table 2-5 and Table 2-6 list the recommended minimum distances between trays, trays and conduits, and conduits, respectively. The dimensions provided in the tables are between the top of one tray/conduit and the bottom of the tray/conduit above, or between sides of adjacent trays/conduits.

Trays containing Data 1 or Data 2 cables should have solid bottoms and covers to complete shielding. Ventilation slots or louvers may be used only in trays containing Data 3 cables.

Continued on next page

2.4 Cabling Considerations, Continued

Spacing, continued

Table 2-4 – Tray spacing

Cable Class	Data 1 mm (in.)	Data 2 mm (in.)	Data 3 mm (in.)	Data 3A mm (in.)	Data 4 mm (in.)	Data 4A mm (in.)
Data 1		Note 2	152.4 (6)	152.4 (6)	660.4 (26)	660.4 (26)
Data 2	Note 2		152.4 (6)	152.4 (6)	457.2 (18)	660.4 (26)
Data 3	152.4 (6)	152.4 (6)		0.0 (0)	Note 1	304.8 (12)
Data 3A	152.4 (6)	152.4 (6)	0.0 (0)		203.2 (8)	457.2 (18)
Data 4	660.4 (26)	457.2 (18)	Note 1	203.2 (8)		0.0 (0)
Data 4A	660.4 (26)	660.4 (26)	304.8 (12)	457.2 (18)	0.0 (0)	

ATTENTION Note the following considerations:

- Note 1:
 - Data 3 and Data 4 cables may be run in a common tray but should be separated by a barrier. This barrier does not have to be grounded.
- Note 2:
 - Data 1 and Data 2 cables may be in a common tray, provided they are separated by a grounded steel barrier.

Table 2-5 – Tray/conduit spacing

Cable Class	Data 1 mm (in.)	Data 2 mm (in.)	Data 3 mm (in.)	Data 3A mm (in.)	Data 4 mm (in.)	Data 4A mm (in.)
Data 1		25.4 (1)	101.6 (4)	101.6 (4)	457.2 (18)	457.2 (18)
Data 2	25.4 (1)		101.6 (4)	101.6 (4)	304.8 (12)	457.2 (18)
Data 3	101.6 (4)	101.6 (4)		0.0 (0)	0.0 (0)	203.2 (8)
Data 3A	101.6 (4)	101.6 (4)	0.0 (0)		152.4 (6)	304.8 (12)
Data 4	457.2 (18)	304.8 (12)	0.0 (0)	152.4 (6)		0.0 (0)
Data 4A	457.2 (18)	457.2 (18)	203.2 (8)	304.8 (12)	0.0 (0)	

Continued on next page

2.4 Cabling Considerations, Continued

Spacing, continued

Table 2-6 - Conduit spacing

Cable Class	Data 1 mm (in.)	Data 2 mm (in.)	Data 3 mm (in.)	Data 3A mm (in.)	Data 4 mm (in.)	Data 4A mm (in.)
Data 1		25.4 (1)	76.2 (3)	76.2 (3)	304.8 (12)	304.8 (12)
Data 2	25.4 (1)		76.2 (3)	76.2 (3)	228.6 (9)	304.8 (12)
Data 3	76.2 (3)	76.2 (3)		0.0 (0)	0.0 (0)	152.4 (6)
Data 3A	76.2 (3)	76.2 (3)	0.0 (0)		152.4 (6)	228.6 (9)
Data 4	304.8 (12)	228.6 (9)	0.0 (0)	152.4 (6)		0.0 (0)
Data 4A	304.8 (12)	304.8 (12)	152.4 (6)	228.6 (9)	0.0 (0)	

Cable, wire and conduit selection

Select the proper cables for each type of system connection. All Programmable Logic Controller (such as the FSC) system wires and cables can be classified into two categories:

- High level conductors
 - used to carry high voltage and/or high current,
 - typically IEEE 518 Data 2, Data 3, and Data 4 categories, and
 - include AC I/O lines; high level DC I/O lines connected to such devices as switches, relays, or solenoids; and AC power lines.
- Low level conductors
 - used to carry low voltage and/or low current,
 - typically IEEE 518 Data 1 category, and
 - include communication cables; DC I/O lines connected to solid state devices such as proximity switches or photoelectric sensors; TTL signal lines; analog signal lines; encoder signal lines; and motion control signal lines.

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2.4 Cabling Considerations, Continued

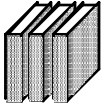
Cabling and wiring guidelines

Observe the guidelines presented in Table 2-7 when routing cables to the FSC Safety Manager cabinet. Pay special attention to separating data cables from power cables to avoid electrical noise problems.

Table 2-7 – Cabling guidelines

Consideration	Description
1	Maintain a 60°C (140°F), or less, ambient air temperature at the bottom of each FSC Safety Manager system.
2	Do all wiring according to IEEE Standard 518-1977.
3	Separate all AC and low-level DC wiring. See Table 2-4, Table 2-5 and Table 2-6 for the recommended minimum distances.
4	Shields of data communications cables must be terminated at both ends. Shields of analog signal cables must be terminated to ground only at the signal source. If a termination is not made, or is intermittent, noise immunity is lost.
5	Data and power cables should cross at right angles and not run in parallel for any distance.
6	Ensure that the entire communications network system is grounded to earth ground.
7	Maintain shield continuity. Ensure that shield leads are not broken.
8	Route cables around rather than through high noise areas.

2.5 Cabling Installation

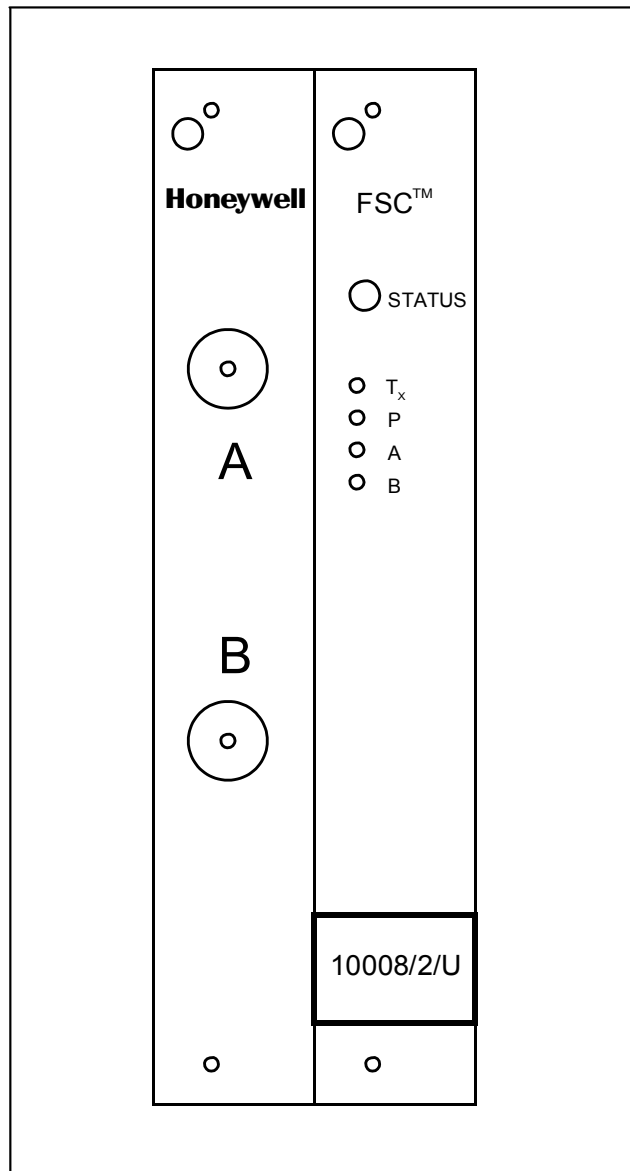


REFERENCE — For information regarding general cabling considerations, refer to Subsection 2.4 — *Cabling Considerations*.

FSC-SMM cabling

Figure 2-1 illustrates the front plate of the FSC Safety Manager Module (FSC-SMM). Connections to the UCN are made via the two UCN F-style connectors in the center of the module (A and B).

Figure 2-1 – FSC-SMM front plate



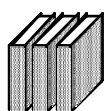
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2.5 Cabling Installation, Continued

FSC-SMM cabling, continued

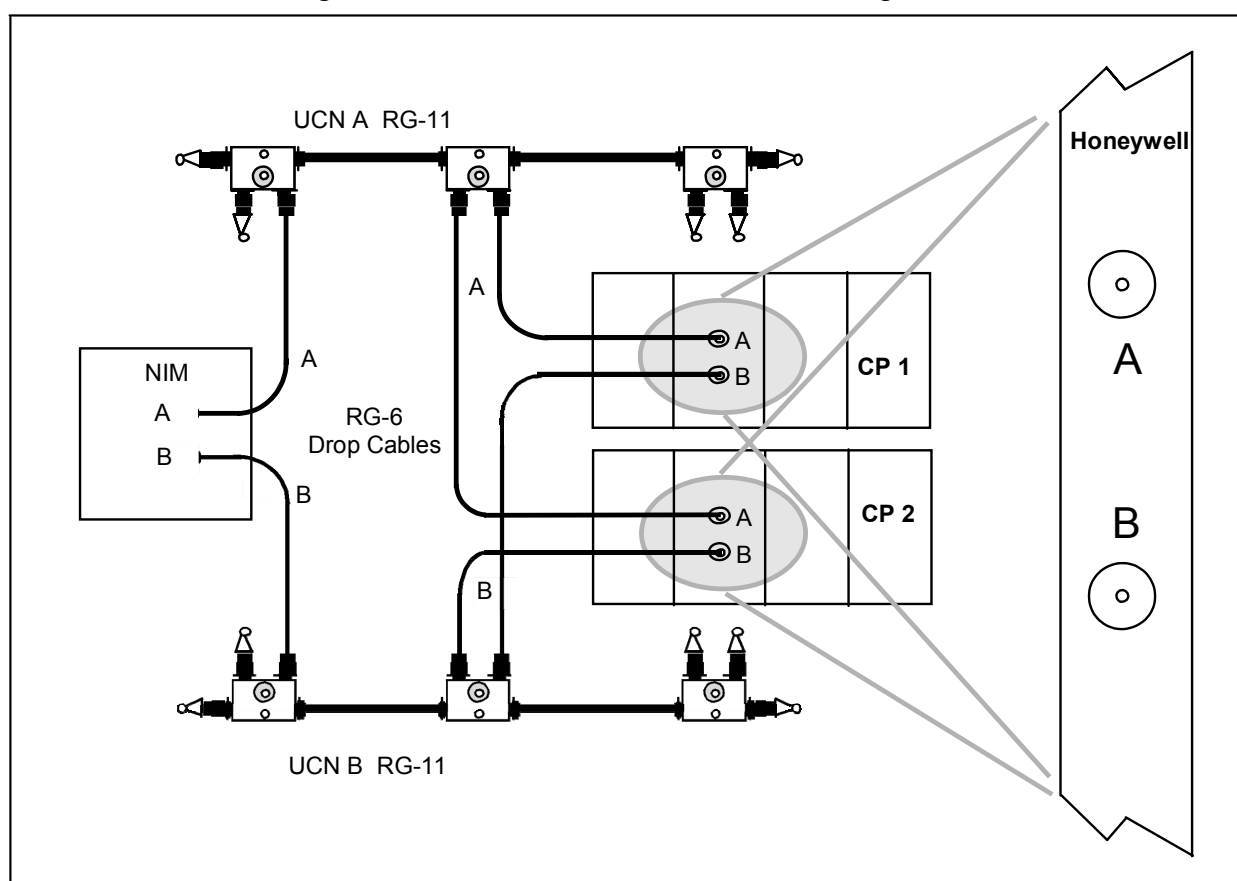
Figure 2-2 illustrates the UCN cabling required of a redundant FSC Safety Manager system.

In this case two FSC-SMM modules are installed, one in each Central Part. RG-6 UCN drop cables are run from the drop ports on UCN A to UCN A F-style connectors of each FSC-SMM. RG-6 UCN drop cables are also run from the drop ports on UCN B to UCN B F-style connectors of each FSC-SMM.



For detailed information on cabling requirements, refer to the *Integration Guidelines for FSC System Cabinets*.

Figure 2-2 – Redundant FSC-SM UCN cabling



CAUTION Remove the UCN cable from the FSC-SMM port without breaking UCN cable continuity, or communication will be lost to other UCN devices.

Continued on next page

2.5 Cabling Installation, Continued

Power supply wiring

ATTENTION For information regarding the wiring of the FSC Safety Manager card file power supply modules, refer to Subsection 2.7 — *System Power Connections*, in this section.

I/O module wiring

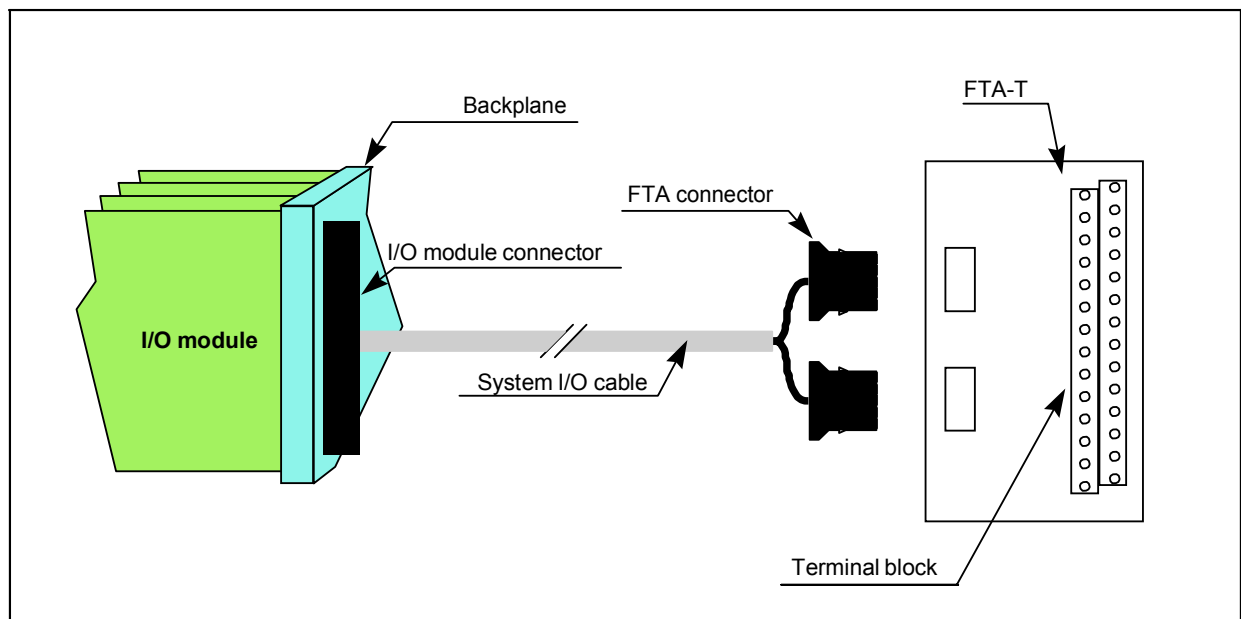
Field wiring is not connected directly to the I/O modules themselves. Rather, as illustrated in Figure 2-3 and Figure 2-4, these connections are made via Field Termination Assemblies (FTAs) available as either an:

- FTA-T (to I/O field cables via terminals), or
- FTA-E (to interface board via system cables).

The FTAs are electrically passive devices which simply pass incoming or outgoing signals between the field and the I/O module.

Figure 2-3 illustrates a typical routing layout of the internal wiring between I/O modules and FTA-T via system interconnection cables (SICs).

Figure 2-3 – Typical routing layout of internal wiring between I/O modules and FTA-T via SICs



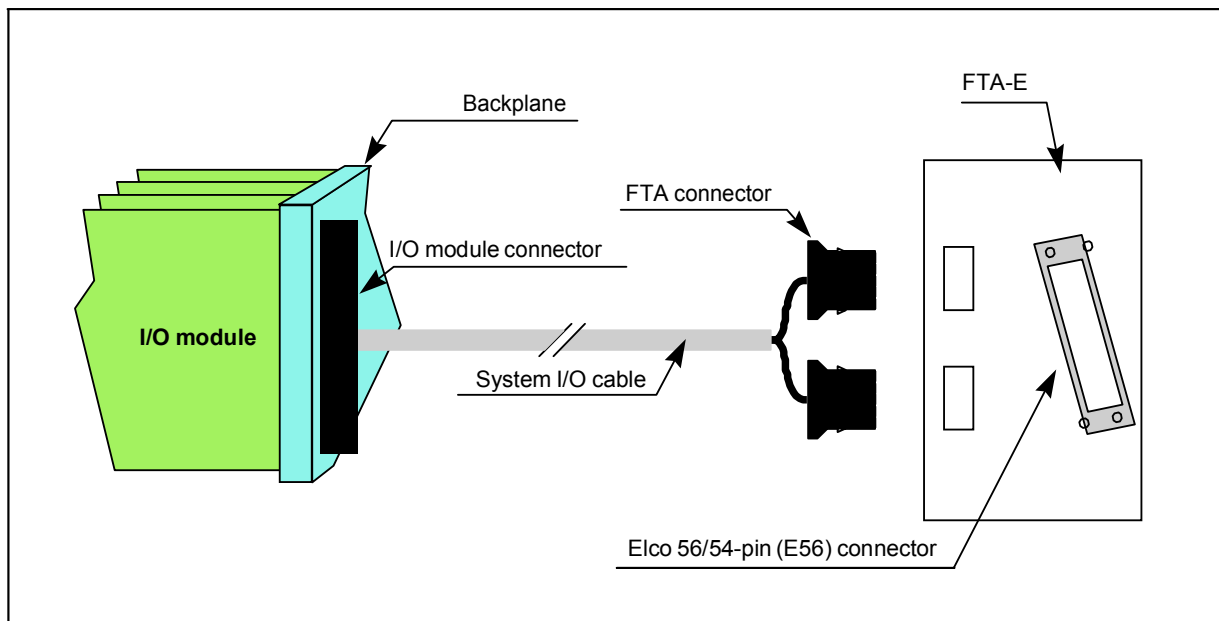
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2.5 Cabling Installation, Continued

I/O module wiring, continued

Figure 2-4 illustrates a typical routing layout of the internal wiring between I/O modules and FTA-E via system interconnection cables (SICs).

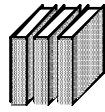
Figure 2-4 – Typical routing layout of internal wiring between I/O modules and FTA-E via SICs



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2.5 Cabling Installation, Continued

Communication module cabling



For more information on communication module cabling, refer to the *FSC Hardware Manual* and the *Integration Guidelines for FSC System Cabinets*.

FSC user station cabling



For more information on FSC user station cabling, refer to the *FSC Hardware Manual*.

2.6 System Power Connections

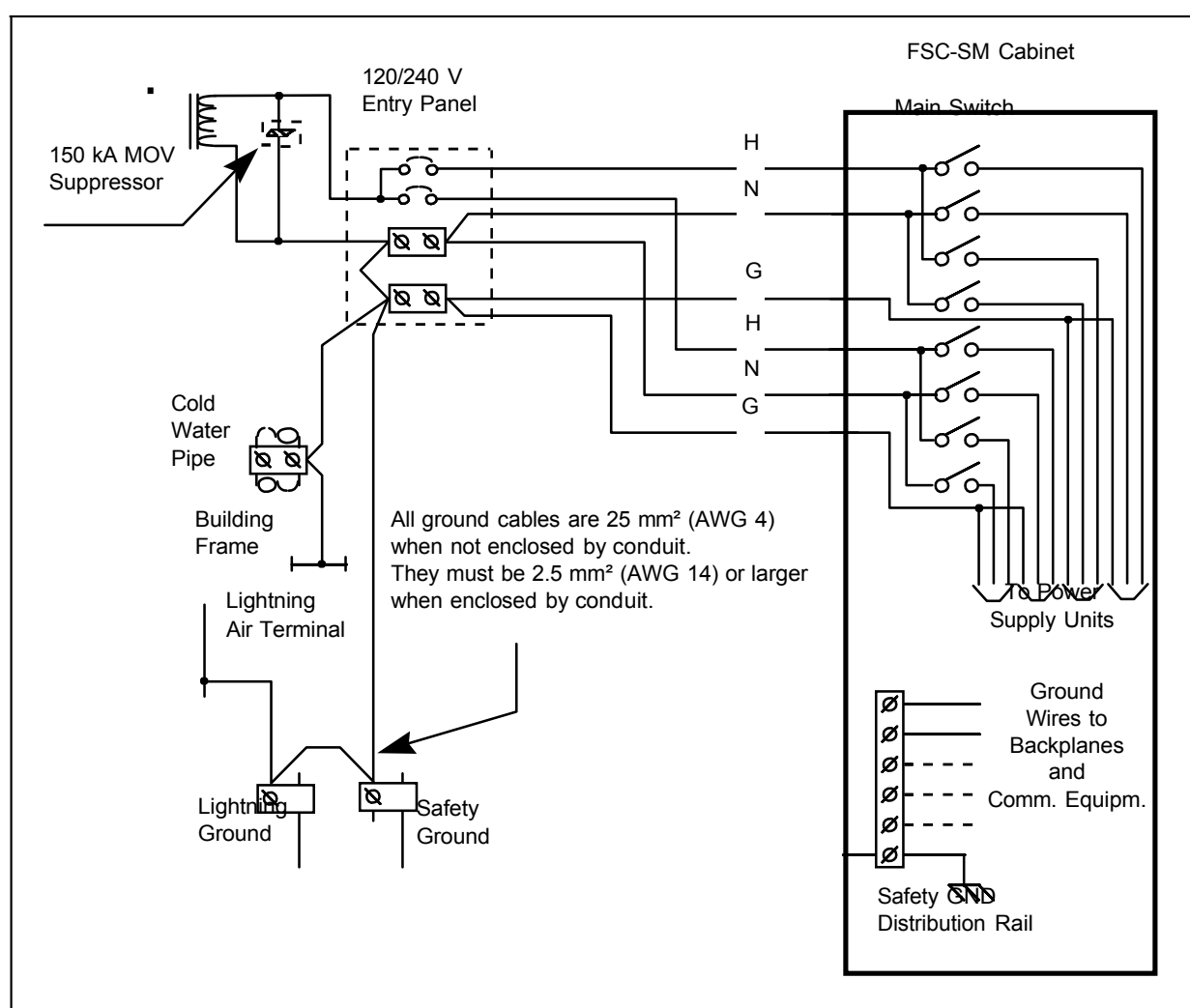
Cabinet power connections

Figure 2-5 illustrates a typical power connection layout for an FSC Safety Manager cabinet.

Conductors and breakers, used to supply power to the cabinet, are sized using rules in the *Integration Guidelines for FSC System Cabinets*. Refer to your site planning drawings and this figure to install power.

A cabinet will have a terminal bar at the bottom of the cabinet to connect two or more feeders.

Figure 2-5 – Typical FSC-SM cabinet power and ground connections



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2.6 System Power Connections, Continued

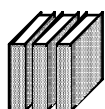
Module power connections

Power is supplied to each FSC Safety Manager Module via the system bus of the Central Part.

2.7 System Ground Connections

Background

The FSC Safety Manager cabinets, backplanes, termination panels, communications equipment and field wiring must be properly grounded to ensure correct and safe operation. This subsection provides the necessary information needed to make these connections.



For basic information regarding the FSC's grounding requirements, refer to *Integration Guidelines for FSC System Cabinets*.

Grounding considerations

Adequate grounding is important for safety considerations and for reducing electromagnetic noise interference. All earth ground connections must be permanent and provide a continuous low impedance path to earth ground for induced noise currents and fault currents. Table 2-8 lists the steps to be performed when grounding your system.

Table 2-8 – System grounding considerations

Step	Action
1	Use the system grounding techniques described in this subsection and those defined in the FSC documentation before making ground connections. Establish: • AC safety ground by making a – temporary connection using the grounding terminal on front of each power supply module, and – permanent, redundant connection to safety ground attaching No. 8 AWG from ground to each card file using a crimped ring lug and 1/4" bolt. • signal ground by tying together the signal ground and safety ground at <u>one</u> and <u>only one</u> point. • shield ground by connecting one end of the shield of each field cable between an analog termination panel and analog field device.
2	Select the proper voltage (115 Vac or 230 Vac) for all configurable modules before supplying power to the system.
3	Securely fasten all cover plates in order to maintain system integrity and to ensure system grounding.

Continued on next page

2.7 System Ground Connections, Continued

Grounding considerations, continued

Table 2-8 – System grounding considerations (continued)

Step	Action
4	Firmly tighten all screws on FSC-SMM module front plates and Central Part coverplates.
5	Consult IEEE Standard 518-1977, the National Electric Code, and Subsections 2.4 and 2.5 of this document before wiring the system components.

Cabinet ground connections

Figure 2-5 illustrates a typical power and ground connections layout for an FSC Safety Manager cabinet.

Ground wires from all mechanical parts and every backplane in the cabinet are collected at a bus bar located at the bottom of the cabinet. This bus bar must be connected to building safety ground with a 2.5 mm² (AWG 14) wire in conduit or a 25 mm² (AWG 4) wire without conduit. Each cabinet must be connected individually to building safety ground and not daisy-chained with other cabinets or equipment.

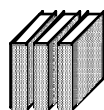
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2.7 System Ground Connections, Continued

Safety ground

The FSC Safety Manager cabinet and external termination panels are properly connected to safety ground in accordance to the specifications and techniques stipulated by FSC.

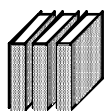
The FSC Safety Manager backplanes and communication equipment are properly connected to safety ground. To insure proper operation of the FSC-SM, safety ground must be configured in accordance with the specifications and techniques stipulated by FSC.



For more information on cabinet grounding, refer to the *Integration Guidelines for FSC System Cabinets*.

Shield ground

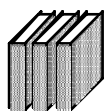
Field wiring between analog field devices for both the internal and external termination panels must be properly connected to shield (earth) ground. To insure proper communication over these links, shield ground must be configured in accordance to the specifications and techniques stipulated by FSC.



For more information on shield grounding, refer to the *Integration Guidelines for FSC System Cabinets*.

Communication equipment grounding

Communication equipment connected to the FSC Safety Manager must be grounded in accordance to the specifications and techniques stipulated by FSC.



For more information on communication equipment grounding, refer to the *Integration Guidelines for FSC System Cabinets*.

2.8 UCN Connections

Node address selection

Unlike other UCN devices, the UCN node address is configured at the FSC user station (using the FSC Development System software and FSC Navigator software), and becomes slot-dependent.

UCN installation

The installation procedure and cabling information can be found in the *UCN Site Planning and Installation* manual in the *UCN Site Planning and Installation* binder.

Figure 2-6 illustrates a typical UCN installation. Connection of the UCN system is made at the front panel of the FSC Safety Manager Module.

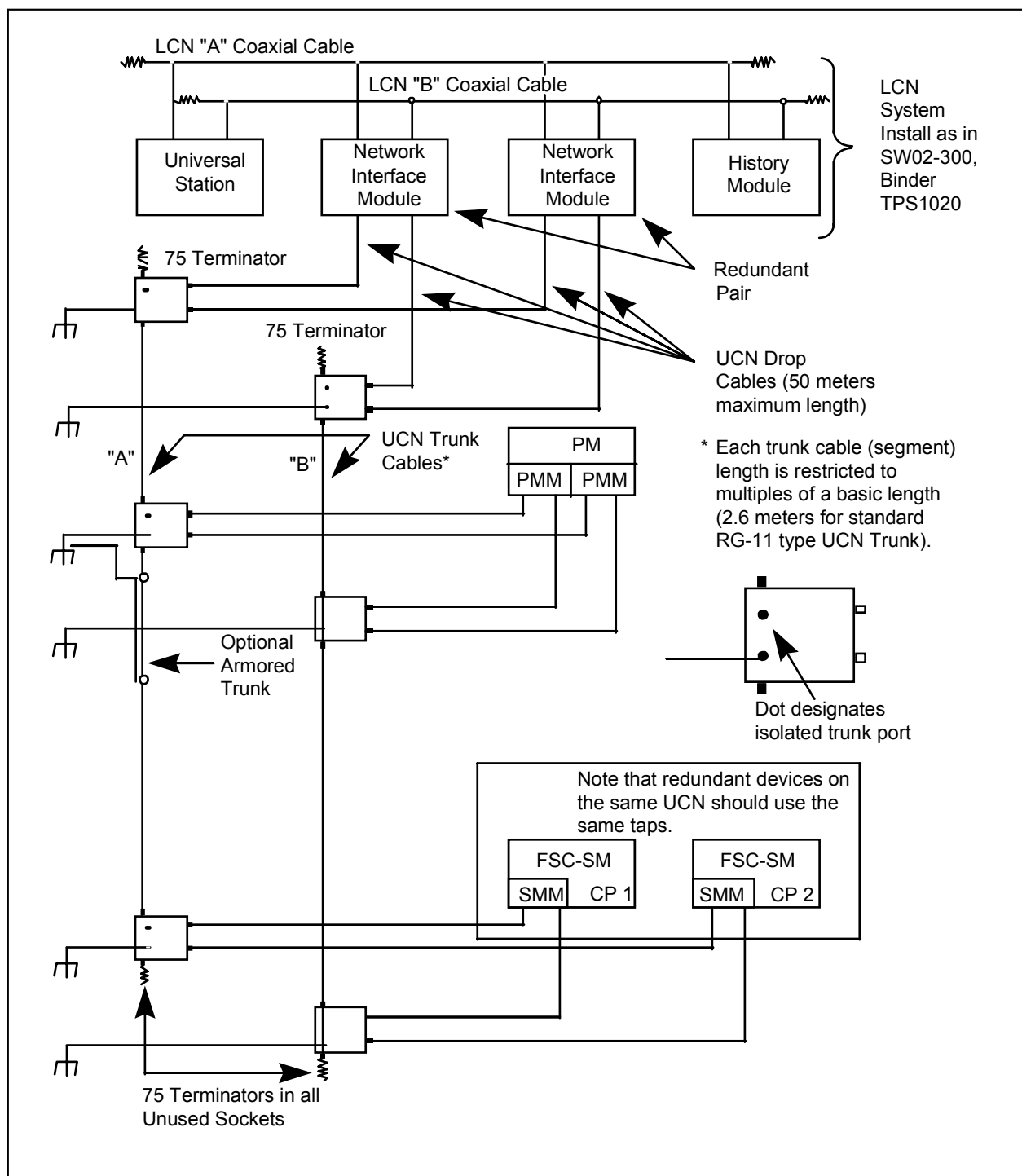
ATTENTION For detailed information regarding the connection of the UCN to the FSC Safety Manager, refer to Subsection 2.5 — *Wiring and Cabling Installation* under the topic, "FSC-SMM Cabling," in this document.

Continued on next page

2.8 UCN Connections, Continued

UCN installation, continued

Figure 2-6 – Connection of the UCN



Continued on next page

2.8 UCN Connections, Continued

Trunk cable connection

Connection of the UCN trunk lines are made at the front panel of the FSC Safety Manager Module. Refer to Figure 2-2 for more details. Use the procedure that is found in the *TPS System Site Planning* manual as the installation guide.

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Section 3 – System Component Configuration

3.1 Section Overview

About this section

This section provides general installation information for various components of the FSC Safety Manager (FSC-SM) system. Topics included in this section are:

Subsection	Topic	See Page
3.1	Section Overview	43
3.2	System Configuration Overview.....	44
3.3	Module Configuration.....	45

ATTENTION This section describes in detail only those installation procedures and considerations specific to the FSC Safety Manager Module (FSC-SMM) and the TotalPlant Solution (TPS) system. Installation procedures and considerations specific to the FSC components can be found in the FSC documentation.

3.2 System Configuration Overview

Introduction

The FSC Safety Manager (FSC-SM) system consists of a redundant Fail Safe Control (FSC) system fitted with a redundant pair of FSC Safety Manager Modules (FSC-SMMs).

Each FSC Central Part contains a central processor unit (CPU), a watchdog (WD) and communication modules (COM). Each Central Part has its own dedicated I/O modules and/or shared I/O modules.

Redundant FSC-SMM modules

The FSC-SM is configured for redundant operation. There can only be one FSC-SMM in each Central Part.

In this configuration, each FSC-SMM is independently connected to the UCN. One of the two will function as the primary, carrying out all UCN to FSC-SM communication and data transfer operations. The other FSC-SMM functions as the secondary or backup link between the UCN and FSC. Database synchronization and status and control transfers between the Primary and Secondary take place over the UCN.

3.3 Module Configuration

Introduction

The FSC Safety Manager is configured at the FSC user station using the FSC Development System software (R42x) or FSC Navigator software (R5xx, R600). There are basically three configuration levels:

- General FSC system configuration,
- Central Part module configuration, and
- I/O module configuration.

Each of these levels is discussed in more detail below.

The starting point for all configuration operations in the FSC Development System software (R42x) is the *Configure FSC system* option from the FSC-DS main menu.

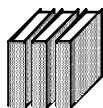
The starting point for all configuration operations in the FSC Navigator software (R5xx, R600) is the *System Configuration* option, which can be called up in three ways:

- Menu bar: Project / System Configuration
- Button bar: I/O
- Navigation area: FSC Project Configuration / System Configuration

General FSC system configuration

The FSC user software (FSC Development System or FSC Navigator) allows the user to configure a number of general system settings such as:

- requirement class,
- FSC system type specification (R42x) or system configuration (R5xx, R600),
- process safety time,
- number of cabinets, and
- number of racks.



For detailed information on the FSC user software and on configuring the FSC system, refer to the *FSC Software Manual*.

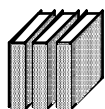
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3.3 Module Configuration, Continued

Central Parts

The Central Parts can only function correctly if they are properly configured, which can be done using the FSC user software (FSC Development System or FSC Navigator). A Central Part (CP) includes several standard items such as the redundant power supplies, the watchdog module and the central processing unit (CPU). It also has a number of free slots in which can be installed:

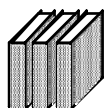
- communication modules (such as the FSC-SMM),
- diagnostic and battery module (DBM), and
- several vertical bus drivers (VBDs).



For detailed information on Central Part requirements and configuration, refer to the *FSC Hardware Manual* and the *FSC Software Manual*.

I/O modules

The allocation of I/O modules within an FSC system and their configuration requirements depend on the safety application and on the FSC system configuration. Use the FSC user software to allocate I/O modules and to enter their safety relation values.



For detailed information on I/O module requirements and configuration, refer to the *FSC Hardware Manual* and the *FSC Software Manual*.

Section 4 – Configuration

4.1 Section Overview

About this section

This section provides general information for various software configuration procedures for the FSC Safety Manager (FSC-SM) system. Topics included in this section are:

Subsection	Topic	See Page
4.1	Section Overview	47
4.2	Configuration at FSC User Station	48
4.3	Configuration at TPS Universal Station	49

Introduction

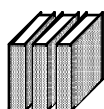
There are three main areas which need to be configured to properly use the FSC Safety Manager within the TotalPlant Solution (TPS) system. These are the following:

- FSC Functional Logic Diagrams (FLDs) programming,
- Configuration of the FSC-SM at the FSC user station, and
- Configuration of FSC-SMM points at the TPS Universal Station (US).

Preconfiguration assumptions

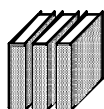
This section assumes the following steps have been completed:

- The FSC user station hardware (i.e. cabling) has been properly installed and connected.



Refer to the wiring section of this manual.

- The FSC user software (FSC Development System or FSC Navigator) has been successfully loaded onto the PC.



Refer to the *FSC Software Manual*.

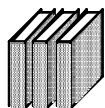
Version needed for configuration

In order to successfully configure the FSC-SM system, you will need to be using release 42x , 5xx or 600 of the FSC software and release 53x or 62x for the TotalPlant Solution (TPS) system.

4.2 Configuration at FSC User Station

Functional Logic Diagrams (FLDs) programming

Functional Logic Diagrams (FLDs) define the control program for the safety application. They are created using the FSC user software: FSC Development System (R42x) or FSC Navigator (R5xx, R600).

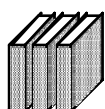


For detailed information on FSC Functional Logic Diagrams programming, refer to the *FSC Software Manual*.

FSC user software configuration

The FSC user software is used to define and configure the safety application. There are several things that need to be configured before the FSC-SM can function properly in the TPS system. The definition and configuration process consists of the following steps:

- defining the hardware configuration of the FSC-SM,
- defining the variables (which includes assigning attributes to them and allocating them to system components),
- allocating system memory for the FSC-SMM,
- assigning a UCN address to the FSC-SMM,
- assigning DCS addresses to the variables,
- compiling the application, and
- transferring the application to the FSC system (either to EPROMs or by downloading it to the system).



For detailed information on the software configuration of FSC-SM, refer to the *FSC Software Manual*.

4.3 Configuration at TPS Universal Station

Introduction

Once the FSC user software configuration has been successfully completed, you will need to use the Universal Station (US) to finish FSC-SM system configuration. This section will discuss the following US configurations:

- Universal Control Network (UCN),
- Node-specific (FSC-SM device), and
- FSC-SMM data points.

Configuration assumptions

This section covers the FSC-SM configuration process at the US. It assumes that you have completed the implementation steps for start-up and have prepared the History Module (HM) and the system for network configuration.

Table 4-1 outlines the start-up process for the TPS system. If these procedures have not been done, refer to the LCN bookset – *Implementation Startup & Reconfiguration*, binder TPS 2030-1.

Table 4-1 – Start-up and reconfiguration procedures

Step	Action	Result
1	Prepare the system for network configuration.	Universal Personality is in "this US". Current time and date is in "this US". The system start-up floppy is in a secure place.
2	Configure the network.	A site-specific Network Configuration File (NCF), required as input to subsequent procedures, is available.
3	Prepare the History Module.	The History Module can now be used as a source for configuration programs and data.

Continued on next page

4.3 Configuration at TPS Universal Station, Continued

UCN configuration

Once you have completed the start-up process for the US, follow the steps outlined in Table 4-2.

Table 4-2 – UCN configuration

Step	Action
1	Press the RESET button. ATTENTION The RESET button is located under the black metal panel at the top of the operator keyboard.
2	Press the LOAD button on the operator keyboard. The system will state 'Loading in progress'.
3	A prompt will appear: "N, 1, 2, 3, 4?". Press N for Network and press the <Enter> key.
4	Another prompt will appear: "OPR, UNP, X?". Press U for Universal Personality and press the <Enter> key.

Continued on next page

4.3 Configuration at TPS Universal Station, Continued

First US display

Once you have loaded the Universal Personality, the System Status display will be the first display to come up. Figure 4-1 shows this display.

Figure 4-1 – US System Status display

MAKE SELECTION										11 Sep 96 14:05:00 3		
SYS VERS: R510					SYSTEM STATUS FROM US13							
NCF VERS: 09Feb95 00:14:19:157												
CABLE A: OK					CABLE B: OK							
US01	US02	US03	US04	US05	US06	US07	US08	US09	US10	US11	US12	US13
OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OK
US14	US15	US16	US17	US18	US19	US20	NM21	NM22	NM23 UCN 02	NM24 UCN 02	NM25	NM26
OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OK	OFF	OFF
NM27	NM28	NM29	NM30	NM31	HG37	HG39	HM43	AM45	AM46	CG 49		
OFF	OFF	OFF	OFF	OFF	OFF	OFF	OK	OFF	OFF	OFF		
LOAD SELECT	LOAD CONFIG	SHOW LOADS	DE- SELECT	NTWK/HWY STATUS	NODE STATUS	STATUS DETAIL	LCN DIAG	TIME/ DATE				
LOAD ISOL	LOAD FAIL	LOAD PWR ON	SELECT AUX INFO	DUMP NODE	SHUT DOWN	LCN OVERVIEW						

Continued on next page

4.3 Configuration at TPS Universal Station, Continued

Engineering Main Menu display

In order to configure your UCN, you will need to bring up the Engineering Main Menu display. To do this, hold the [CTRL] key on the engineering keyboard and press the [HELP(M)] key. Figure 4-2 shows this display.

Once you have brought this display up, select the NETWORK INTERFACE MODULE target.

Figure 4-2 – US Engineering Main Menu display

11 Jun 97 14:05:00 3				
ENGINEERING MAIN MENU				
UNIT NAMES	HIWAY GATEWAY	PICTURE EDITOR		
AREA NAMES	LOGIC BLOCKS	FREE FORMAT LOGS		
CONSOLE NAMES	APPLICATION MODULE	BUTTON CONFIGURATION		
LCN NODES SYSTEM WIDE	COMPUTING MODULE	HM HISTORY GROUPS		
VALUES VOLUME	NETWORK INTERFACE MODULE			
CONFIGURATION				
		DOCUMENTATION TOOL		
	BUILDER COMMANDS	AREA DATA BASE		
Support Functions and Utility Programs				
COMMAND PROCESSOR	SYSTEM MENU	SYSTEM STATUS	CONSOLE STATUS	
SUPPORT UTILITIES	SMCC/ MAINTENANCE			
R530 (C) HONEYWELL INC., 1984-97				

Continued on next page

4.3 Configuration at TPS Universal Station, Continued

NIM Build Type Select Menu display

The NIM Build Type Select Menu display will be the next display to come up. This display is shown in Figure 4-3. Select the UCN NODE CONFIGURATION target to configure the UCN.

Figure 4-3 – US NIM Build Type Select Menu

MENU PAGE 01 OF 01

NIM BUILD TYPE SELECT MENU

UCN NODE CONFIGURATION

NODE SPECIFIC CONFIGURATION

PROCESS POINT BUILDING

LIBRARY CONFIGURATION

F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

UCN Node Configuration display

You should now be in the UCN Node Configuration display. Figure 4-4 shows this display. Do the steps in Table 4-3 to configure the UCN.

Table 4-3 – UCN node configuration

Step	Action
1	Enter the requested data in the given slots. All configuration information should be taken from the FSC-SM Configuration Forms you have filled out.
2	Select the FSC-SM for UCN node type and FSC for the FSC-SM platform.
3	Press the [F12 LOAD] key to load your configuration.

ATTENTION The node number you select must correspond with the node number you entered for the UCN configuration at the FSC user station.

Continued on next page

4.3 Configuration at TPS Universal Station, Continued

UCN Node Configuration display, continued

Note that the network number can be from 1 to 20. The node number can range from 1 to 63, using odd numbers only.

Figure 4-4 – US UCN Node Configuration display

11 Jun 97 14:05:00 3					
PED >>>>>> POINT:\$NM02N13		UNIT:SY		PAGE 01 OF 01	
UCN NODE CONFIGURATION					
NETWORK NUMBER	(NTWKNUM)	02			
NODE NUMBER	(NODENUM)	13			
UCN NODE TYPE	(NODETYP)	NOTCONFIG	NIM	PM	APM
		HPM	LM	SM	
SM PLATFORM	(SMPLTFM)	TRICON	FSC		
NODE ASSIGNMENT	(NODEASSN)	THISNIM	REMOTNIM		
DIGITAL COMPOSITE PV STATE TEXT					
MOVING TEXT	(MOVPTXT)	MOVING			
BAD TEXT	(BADPTXT)	BAD			
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=					
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD					

Continued on next page

4.3 Configuration at TPS Universal Station, Continued

Node-specific configuration

To configure the node-specific data, return to the NIM Build Type Select Menu by pressing the [F9 WLK BACK] key (refer back to Figure 4-3). Select the NODE SPECIFIC CONFIGURATION target. The Node-Specific Configuration display is shown in Figure 4-5.

Enter and select the desired configuration. Press the [F12 LOAD] key.

ATTENTION Make sure the network and node numbers entered for this display match those entered in the UCN Configuration display.

There is no need to configure the redundant FSC-SM device. All FSC-SM-related information will be automatically transferred to the "new" primary FSC-SM should there be a failover or switchover to the backup.

Figure 4-5 – US Node-Specific Configuration display

11 Jun 97 14:05:00 3

PED >>>>>> POINT:\$NM02B13UNIT:SYUNENTEREDPAGE 01 OF 03

NODE-SPECIFIC CONFIGURATION

NETWORK NUMBER	(NTWKNUM)	02		
NODE NUMBER	(NODENUM)	13		
NODE TYPE	(NODETYP)	PM	APM	HPM
		LM		
SM PLATFORM	(SMPLTFM)	TRICON	FSC	

*
* WARNING: IF THE POINT MIX OR THE SCAN
* RATE IS CHANGED, THEN THE NODE'S FULL
* DATA BASE WILL BE SET TO ITS DEFAULT.
* THE DISPLAY LINES WHICH CAN CAUSE THE
* DATA BASE TO BE DEFAULTED ARE STARRED
* AND HAVE AN ASTERISK (*) IN COLUMN 80.
*

F1=PEDF3=F5=OVERWRITEF7=RECONF9 =WLK BACKF11=

F2=RECALL DISPF4=F6=F8=PED STATUSF10=WRITEF12=LOAD

Continued on next page

4.3 Configuration at TPS Universal Station, Continued

Process Point Building display

You must once again return to the NIM Build Type Select Menu to access the FSC-SMM data point building displays. To do this, press the [F1 PED] key.

Select the PROCESS POINT BUILDING target, once you have returned to the NIM Build Type Select Menu. Figure 4-6 shows the display which will be brought up.

Figure 4-6 – US Process Point Building display

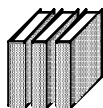
11 Jun 97 14:05:00 3					
MENU		PAGE 01 OF 01			
NIM BUILD TYPE SELECT MENU					
NIM PROCESS POINT BUILD TYPE					
		ANALOG INPUT	ANALOG OUTPUT		
		DIGITAL INPUT	DIGITAL OUTPUT	DIGITAL COMPOSITE	
		REGULATORY PV	REGULATORY CONTROL		
		LOGIC			
		FLAG	TIMER	NUMERIC	
		ARRAY	DEVICE CONTROL		
		PROCESS MODULE			
F1=PED F3= F5=OVERWRITE F7=RECON F9 =WLK BACK F11=					
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD					

Continued on next page

4.3 Configuration at TPS Universal Station, Continued

Building FSC-SMM data points

This subsection takes you through the general steps for FSC-SMM data point configuration. To build FSC-SMM data points, you will need to follow the steps listed in Table 4-4.



For more information refer to *Section 4 — Data Point Detailed Description* in the *FSC Safety Manager Control Functions*.

ATTENTION Information on the definition of a specific parameter can be provided to you online by pressing the [HELP] key for the parameter you have selected.

Table 4-4 – FSC-SMM data point configuration procedure

Step	Action
1	Once you are in the Process Point Building display, select the FSC-SMM point type you wish to configure.
2	Enter and select the desired configuration scheme. Use the [PAGE FWD] button on the operator keyboard to move to the next configuration page as needed. ATTENTION Data entered should be based on the FSC-SM Configuration Forms you have filled out.
3	Press the [F12 LOAD] key to load the data in the FSC-SM database.
4	Press the [F9 WLK BACK] key to return to the Process Point Building display.
5	Repeat steps 1 to 4 for the next FSC-SMM data point type until all point types are configured.

ATTENTION FSC aliases must be configured before points can be downloaded against given aliases.

Left blank intentionally

Index

A, B

Bolt-down, *19*
Bolt-down holes, *6*
Building FSC-SMM data points, *57*

C

Cabinet entry, *7, 8, 9, 10, 11, 12*
Cabinet ground connections, *37*
Cabinet power connections, *34*
Cable conduits, *24*
Cable entry to cabinet, *7, 8, 9, 10, 11, 12*
Cable trays, *24*
Cable, wire and conduit selection, *27*
Cabling
 Communication module, *33*
 FSC user station, *33*
 FSC-SMM, *29*
 Redundant, *30*
Cabling guidelines, *28*
Central Parts, *46*
Communication equipment grounding, *38*
Communication module cabling, *33*
Conductors
 High level, *27*
 Low level, *27*
Conduit spacing, *27*
Conduits, *24*
Configuration
 Central Parts, *46*
 FSC Development System (FSC-DS), *48*
 FSC Navigator, *48*
 General, *45*
 I/O modules, *46*
 Universal Station (US), *49*
Configuration of FSC Safety Manager, *45, 47*
Crane, *19*

D

Data points, *57*
Data signal cable, *25*

E

Engineering Main Menu display, *52*
Environmental conditions, *2*

F

Field termination assemblies
 FTA-E, *7*
 FTA-T, *7*
Fork lift, *14*
FSC Development System (FSC-DS), *45, 48*
FSC Navigator, *45, 48*
FSC user station cabling, *33*
FSC-SMM cabling, *29*
 Redundant, *30*
FSC-SMM data points, *57*
Functional Logic Diagrams (FLDs), *48*

G

General FSC system configuration, *45*
Grounding
 Cabinet connections, *37*
 Communication equipment, *38*
 Safety, *38*
 Shield, *38*
 Signal, *38*
Grounding considerations, *36*

H

High level conductors, *27*
History Module, *49*

I, J, K

I/O module wiring, *31, 32*
I/O modules, *46*
Installation checklist, *22*

Index

L

Leveling, *19*
Low level conductors, *27*

M

Mobile lifter, *15*
Module configuration, *45*
Module power connections, *35*
Move checklist
 Electronics room, *5*
 FSC-SM cabinets, *4*
Moving FSC-SM cabinets, *3, 4, 5, 13*
Moving the equipment
 Crane method, *19*
 Fork lift, *14*
 Mobile lifter, *15*
 Roller method, *18*

N, O

Network configuration, *49*
NIM Build Type Select Menu display, *53*
Node-specific configuration, *55*
Noise generation, *25*
Noise susceptibility, *25*

P, Q

Power supply wiring, *31*
Process Point Building display, *56*

R

Redundant cabling, *30*
Redundant FSC-SMM modules, *44*
Relative humidity, *2*
Roller, *18*

S

Safety ground, *38*
Shield ground, *38*
Shielding, *24*
Shock, *2*
Software
 FSC Development System (FSC-DS), *45, 48*
 FSC Navigator, *45, 48*
Spacing
 Conduits, *27*
 Trays, *26*
 Trays/conduits, *26*
Storage ratings, *2*
System configuration, *44*
System status display, *51*

T

Temperature, *2*
Test equipment, *1*
Tools, *1*
Tray spacing, *26*
Tray/conduit spacing, *26*
Trunk cable connection, *41*

U

UCN installation, *39*
UCN Node Configuration display, *53*
Universal Personality, *51*
Universal Station (US), *49*
Unpacking, *13*
US configuration, *49*

V

Version needed for configuration, *47*
Vibration, *2*

W, X, Y, Z

Wiring classifications, *25*
Wiring guidelines, *28*

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